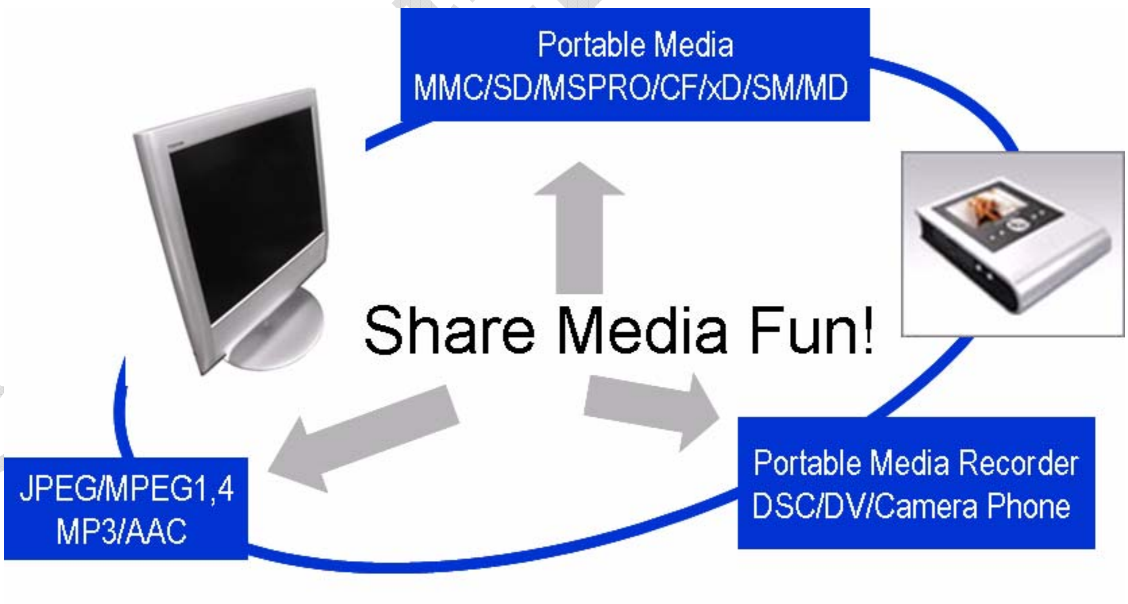
Media Player Controller

MP612

Datasheet

Revision 0.81

2005/03/30



Magic Pixel Inc.

TABLE OF CONTENT

1 General Description

The MP612 is a general-purpose LSI for digital media player application. It implements all necessary functions in hardware, but also allows much flexibility for customization. It provides a variety of choices for internal processing and peripheral configuration. Customer can always choose the best combinations to fit their own application requirement.

The central part of the chip is a high performance 32-bit RISC CPU which allows flexible system control. It contains 2KB of Instruction Cache and 2KB of Data Cache to save the memory fetch overhead and hence boost the software performance further.

A flexible image scaling architecture is implemented to allow image resizing in different operating modes.

A versatile audio interface is implemented so that standard AC97 or non-standard audio CODEC can be easily incorporated. With the video & audio processing capabilities, **the user can playback most of popular digital media, including still picture (JPEG/ BMP/ TIFF), digital audio (MPEG-1 Level 1, 2/MP3/ AAC_LC/AMR) and Video (Motion JPEG /MPEG1/MPEG4) for A/V entertainment device.**

The Image Display Unit *(IDU)* is implemented to allow image display through digital LCD or TV. **It builds in the high performance TV Encoder that supports both Analog SDTV and HDTV video formats (Up to D4, i.e. 1080i, resolution)**. A scaling unit is included for scan conversion. It also supports bitmap On-Screen Display. Text or graphic messages can be overlaid with the picture for friendly user interface.

During Playback, the compressed picture is decompressed and sent to IDU for image display. It includes a JPEG engine to provide high performance image compression and decompression. **Up to VGA@30fps of video clip playback can be achieved.**

MP612 provides a versatile interface to support most of popular memory card standards, including Compact Flash, Smart Media, PC-ATA, MMC, SD, xD, Memory Stick and IBM Micro Drive. It offers the maximum flexibility to the customer.

For data transfer to/from PC, it provides both USB and popular UART interfaces. **It includes a USB 2.0 device controller that is compliant with the USB 2.0 standard**. 4 endpoints have been implemented to achieve variety of requirements for image/audio data upload or download. The built-in UART controller supports up to 1Mbps of baud rate. It also includes a IR(Infra-Red) controller supporting NEC Button and Remote Point Mouse protocols. Besides, it also includes a flexible parallel peripheral interface to expand the application functions, like IEEE1394, Bluetooth, WLAN, with external controllers.

For flexible peripheral control and user interface, it provides several GPIOs, including 4 PWM outputs. The memory interface is designed to support SDRAM/ROM/Flash/SRAM in most flexible combinations.

With the abundant features and superior performance and quality, MP612 provides an excellent solution for digital photo frame, and media player application.

2 Features

- **Power**
  - Dual Power. 1.8V for core, and 3.3V/1.8V for I/O
  - Advanced power saving control
- **Image Processing Unit**
  - Hardware scaling engine to scale up and down images for resolution conversion or image zooming
- **JPEG Codec**
  - Support image resolution up to 8Kx8K
    - High speed JPEG compression and decompression (48MPixel/sec)
  - Support image sub-sampling after JPEG decompression
  - Support image rotation
  - Restore JPEG image after adjusted to memory card
- **Audio/Video decoding accelerator**
  - Support AAC_LC, AMR, MPEG-1 Layer 1, 2 and MP3 decoding.
  - Hardware accelerator for MPEG-1 and MPEG-4/H.263 video decoding
  - Hardware accelerator for MPEG-1 Layer 1, 2, MP3 and AAC-LC decoding
- **Memory Interface**
  - Support SDRAM from 16Mb up to 256Mb x16-bit /32-bit, x1 or x2
  - Support DDR SDRAM 16-bit data bus
  - Support NAND-type Flash memory (8Mb and above) x1 or x2
    - Support NOR-type Flash memory, up to 64MB with x8 or x16-bit data width for both BIOS access and data storage
    - Support program boot up from NAND flash memory
- **Memory Card and Hard drive Interface**
  - Support Smart Media, Compact Flash, Multi-Media Card (MMC, MMC4.0/3.2), Security Disk (SD), Memory Stick Pro, xD Picture Card(1.0/1.1), and IBM Micro Drive
  - Support parallel ATA with PIO, Multi-DMA and Ultra DMA-66 modes
  - Support multi-sector DMA
    - Support card-to-card and card-to/from-ATA copy.
- **Audio Interface**
  - Support 3.3V 8-bit and 16-bit voice/audio CODEC for voice/audio recording and playback
  - Support AC-97 CODEC interface
  - Support I2S interface
  - Support synchronized voice and video stream playback
- **USB Interface**
  - High-speed USB 2.0 device function with embedded USB PHY

- Embed USB1.1 Host Controller to connect to pen drive
- Support Direct Print function (Pict-Bridge)

- Power saving control to comply with USB spec.
- Support uploading and downloading capability
- Support USB Mass Storage Class for both High Speed Device and Full Speed Host functions

- **Display Interface**
- **Analog Output**
  - Support analog YPbPr/YCbCr outputs up to D4 resolution (1920x1080i @60Hz).
  - Support analog RGB output up to WXGA@50Hz (1366x768) / XGA @60Hz (1024x768).
  - Support CVBS and S-Video output.
- **Digital Output**
  - Support digital YUV data output conforming to ITU-R BT6.01/656/709 up to D4 resolution (1920x1080i)
  - Support digital RGB888 data output up to WXGA @50Hz (1366x768) / XGA @60Hz (1024x768) or down frame rate for higher resolution output
  - Support various kinds of digital-interface of small size LCD displays of Epson, AU, Casio, SONY, Sanyo, and etc.
- **Colorful Graphic OSD**
  - Support 2-, 4- or 8-bit palette-indexed Bit-mapped On Screen Display(OSD) for content-rich display user-interface
- **CPU & Misc.**
  - Embedded high performance 32-bit RISC processor with 2KB instruction cache and 2KB data cache and 512MB memory addressing capability
  - BIOS storage in Flash memory with in-system-programming (ISP) capability
  - Embedded phase-locked loop(PLL) with independently programmable multiple clock outputs
  - Support up to 96MHz system clock
  - Flexible GPIO control for variety of peripheral control, like monochrome LCD driver IC, button control, melody IC, buzzer, and etc.
  - All pins can be re-configured as GPIO pins when their primary-defined functions are not in use
- **Package**
  - 256-ball LFBGA (17mm x 17mm x 1.7mm, Ball Pitch = 1.0mm)
- **System Development Kit & Software Support**
  - Support real time-OS

  - U-Itron
- Firmware and driver support

  Windows 98, Windows ME, Windows 2000, Windows NT 4.0/5.0 , Windows XP, Linux, and iMac drivers
- Schematics and application note

3 Block diagram

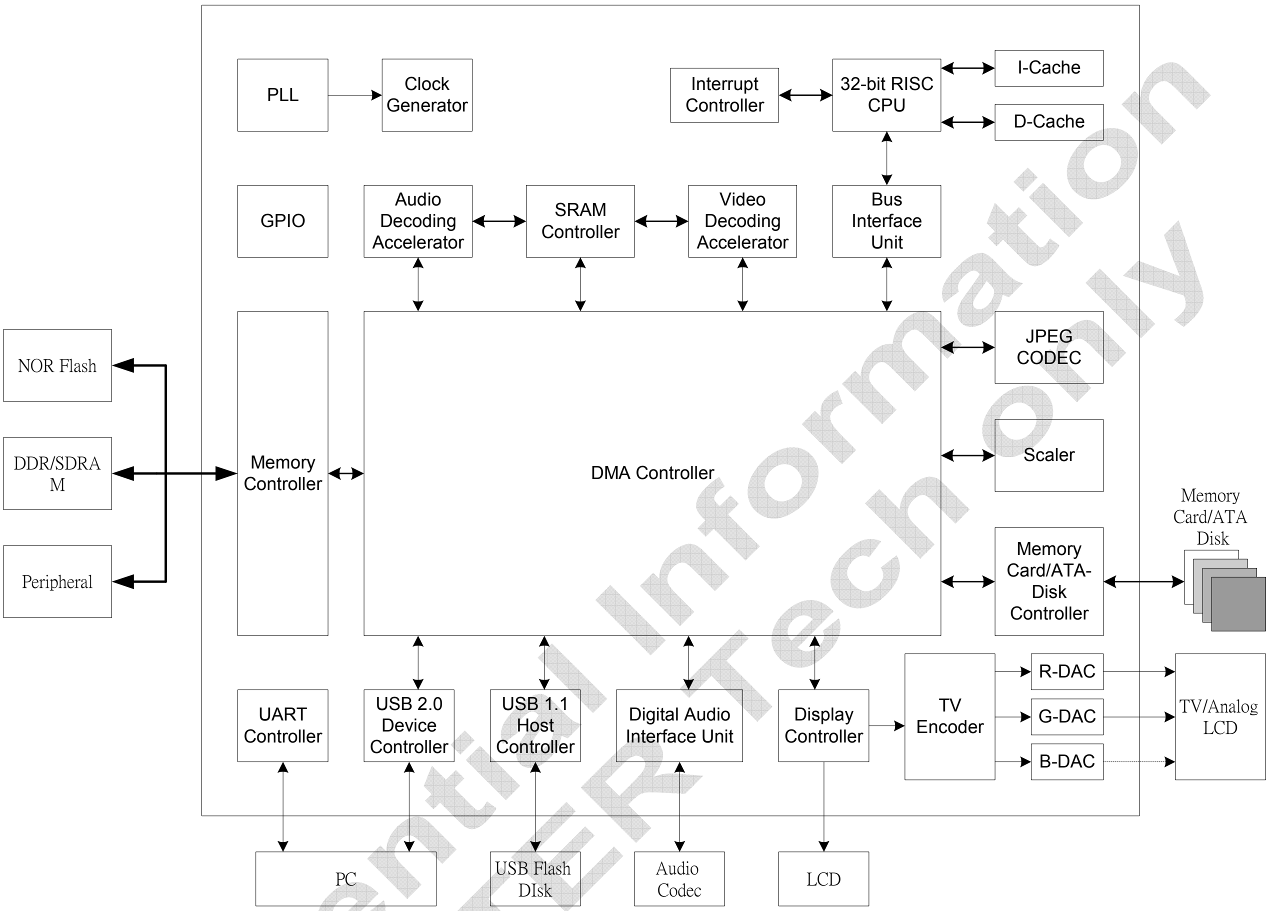


Figure 3-1 MP612 Block Diagram

4 Functional Description

The MP612 is a general-purpose LSI for digital media player application. It implements all necessary functions in hardware, but also allows much flexibility for customization. It provides a variety of choices for internal processing and peripheral configuration. Customer can always choose the best combinations to fit their own application requirement.

4.1 Image Processing Unit (IPU)

Because the images captured by digital still cameras may differ in resolution to the display device's resolution. To be capable of full screen display and image zoom in function, the MP612 embeds a flexible scaling engine. The core of image resizing is a bi-linear scaler and two image sub-sampling blocks. Image shrinking and enlargement with feasible quality are both supported. By applying multi-pass image scaling, the allowable processing resolution is unlimited.

4.2 Image Display Unit (IDU)

The Image Display Unit supports a variety of digital LCD interface, including small size panels of AU, Epson, Casio, SONY and Sanyo. Analog LCD can also be supported with the proprietary Serial DAC through the Serial DAC interface. It also embeds the high performance TV Encoder that supports both Analog SDTV and HDTV video formats (Up to D4 resolution). Besides, it also supports digital video outputs as ITU. BT.601 , BT.656, and BT.709 up to D4 resolution(1920x1080i). A programmable color space conversion unit is also implemented to support both RGB and YCbCr/YPbPr outputs.

MP612 provides great flexibility in display timing settings. Typical display output timing is illustrated in Fig. 4-2-1 and Fig. 4-2-2. All the parameters shown in the figures are programmable.

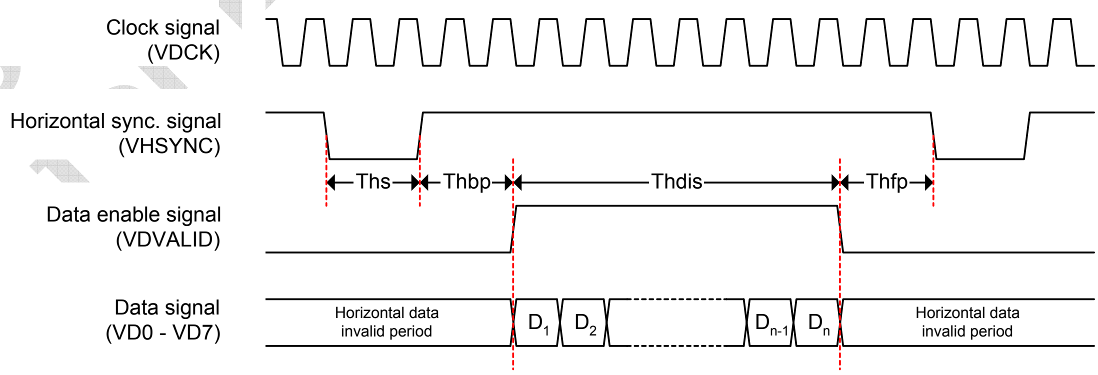


Fig. 4-2-1. Display output horizontal timing diagram

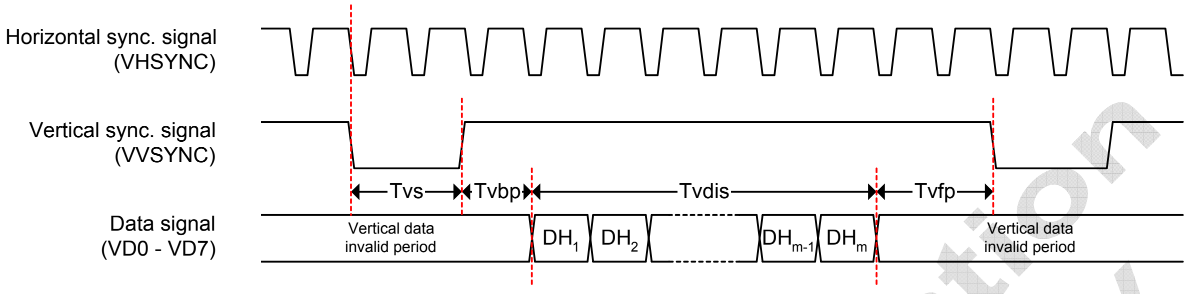


Fig. 4-2-2. Display output vertical timing diagram

A bitmap OSD is also included. 8 , 4 or 2 bits index per pixel can be chosen. With 8-bit index, 128 OSD colors and 8 programmable blending colors can be selected. Or with 2-bit index, only 3 programmable blending colors can be used. But the latter case will save the required memory size of the OSD bitmap. Through the Palette registers that specify the OSD color and blending ratio, it allows quite flexible and content-rich user interface.

4.3 JPEG CODEC

A standard JPEG Compression/Decompression engine is embedded. It is compliant to the Baseline JPEG Standard. It accepts image data in YCbCr 4:2:2 or 4:4:4 format. YCbCr 4:2:0 compression format is also supported, but the source image in DRAM buffer should be in 4:2:2 format still. CDU will perform the sub-sampling on Cb and Cr.

	4:2:0	**4:2:2**	**4:4:4**
MCU Types	8, 8: Y Y / Y Y, Cb, Cr	8, 8: Y Y, Cb, Cr	8, 8: Y, Cb, Cr
MCU size	16 × 16	16 × 8	8 × 8

Table 4-3-1. MCU types supported by CDU

It is capable of processing image with resolution up to 4Kx4K. The image source and compressed stream are both acquired from and delivered to SDRAM through DMA. It needs no CPU intervention for data transfer. Single cycle throughput can be achieved to allow real-time capturing. It can always catch up and deliver high performance for continuous image or video capturing.

It supports two sets of Quantization and Huffman tables respectively. They are both programmable for better compression quality adjustment. The latter is usually fixed in factory, while the Quantization table can be adjusted with pre-determined values for image quality and compression ratio tradeoff controlled by the end users.

It also supports Restart Marker insertion and detection when enabled. Other markers or headers should be handled by the firmware instead.

The JPEG module consists of System Bus Interface, configuration registers, dedicated SRAM modules, system memory interface, Scaling Unit and Control Unit.

CPU can configure and start/stop encoding/decoding process via the 32-bit System Bus Interface.

In encoding mode, it reads in images pixel data, from external system SDRAM and output the encoded stream data back to SDRAM.

While decoding, stream data is read from SDRAM, with header information (Huffman tables, quantization tables, etc.) already peeled by CPU. Pixel data is output MCU by MCU.

An interrupt is triggered when encoding/decoding process is finished.

4.4 MPEG-1,4 Decode

The MPEG-1,4 decoder performs video decoding process and most of the tasks are implemented by hardware. The motion compensation (MC), IDCT, inverse quantization, and AC/DC prediction are all implemented by hardware to minimize the power consumption and to reduce the load on system CPU. This MPEG-4 decoder supports MPEG-4 advanced simple profile* decoding with resolution up to VGA (640x480).

** Except interlaced mode and global motion estimation support.*

- **Main Features**

  The main features of the MPEG-1,4 Decoder are summarized as below.

  - Full features of MPEG-4 Simple file, and partial Advanced Simple Profile except interlace-scan and Global Motion Compensation (GMC)
  - Resolution up to VGA (640x480)
  - Real-time decoding with frame rate up to 30 fps
  - All MPEG-4 error resilience tools
  - Supports MPEG-4 4 MV
  - Unrestricted motion vector value
  - Post-processing (hardware de-blocking)
  - Error detection and concealment function

4.5 AAC / MP3 Audio Decoder

Under the existing DPF architecture, the MP3+AAC Audio decoder will be implemented partially using HW to accelerate the decode function. The implemented MP3 decoder, in conformance with the ISO/IEC 11172-3 specification, will also decode layer1 and layer2 audio, the layer 1 and 2 are fully decoded using SW, while the AAC decoder will conform to ISO/IEC 13818-7 and ISO/IEC 14492-3 Low Complexity.

- **Overview**

The block diagram below shows all the components of the MP3+AAC decoder, the bit-stream is first parsed by SW for synchronization (frame detection) and optional CRC check. The bit-stream is partitioned into frames and granules for MPEG1, each layer1/2 frame contains all required information to reconstruct the audio samples within the frame, but for layer 3 and AAC, previous frame information are needed to reconstruct current frame. After Huffman decoding, SW will transfer VLC decoded data to external memory for storage and will continue process layer 1 and 2 data. For Layer 3 and AAC, SW will send the data to the Requantization block through the DMA interface, after the decoded samples are re-scaled by the Requantization block, the new data are stored in a temporary buffer. For stereo decoding both left and right channel data must be buffered, thus the temp buffer must be able to hold two channels of data. After stereo decoding, the stereo-decoded samples are written back to the temporary buffer. The stereo decoding function is followed by IMDCT(including windowing and block-switching) to transform the frequency samples back to time-domain, the resulting time samples will be saved back to the temp-buffer. As the buffer still contains original frequency samples, there will be associated buffering to hold the results such that the result will not overwrite the frequency samples that are still needed. The IMDCT will produce the final PCM samples for AAC coded streams, but for MP3, an extra synthesis filterbank will be needed to convert MP3 data back to PCM format. DMA interface will send the PCM back to system memory.

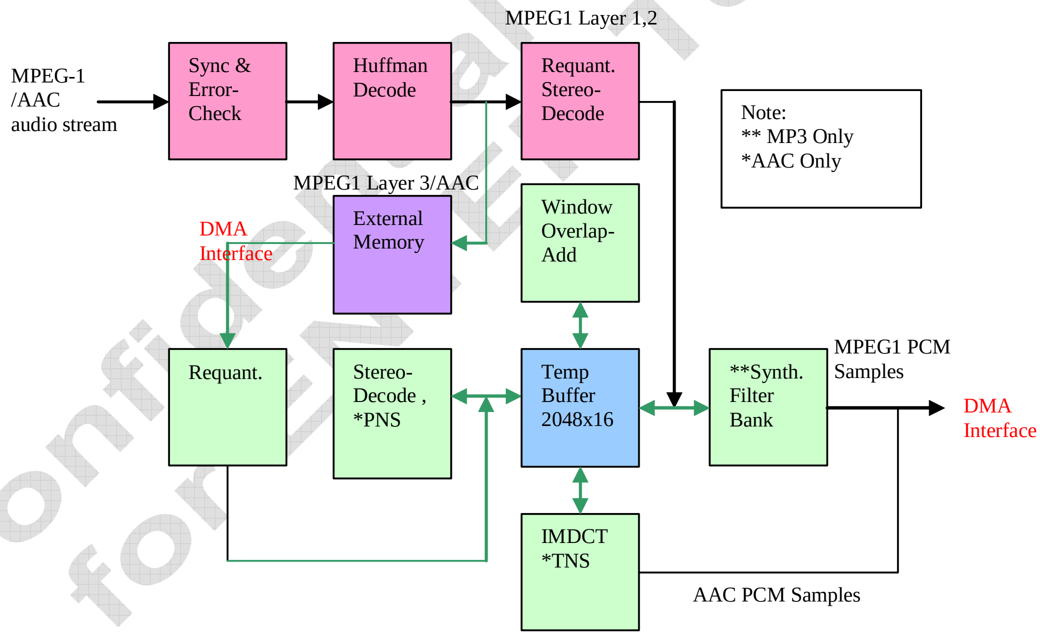


Figure 4-4-1: MPEG-1 Audio Decoder Functional Diagram

Figure 4-4-2 shows the processing pipeline for the MP3 decoder, the flow is similar for AAC, except the addition of PNA and TNS processing stages and no sub-band processing as well as the lack of final synthesis filterbank.

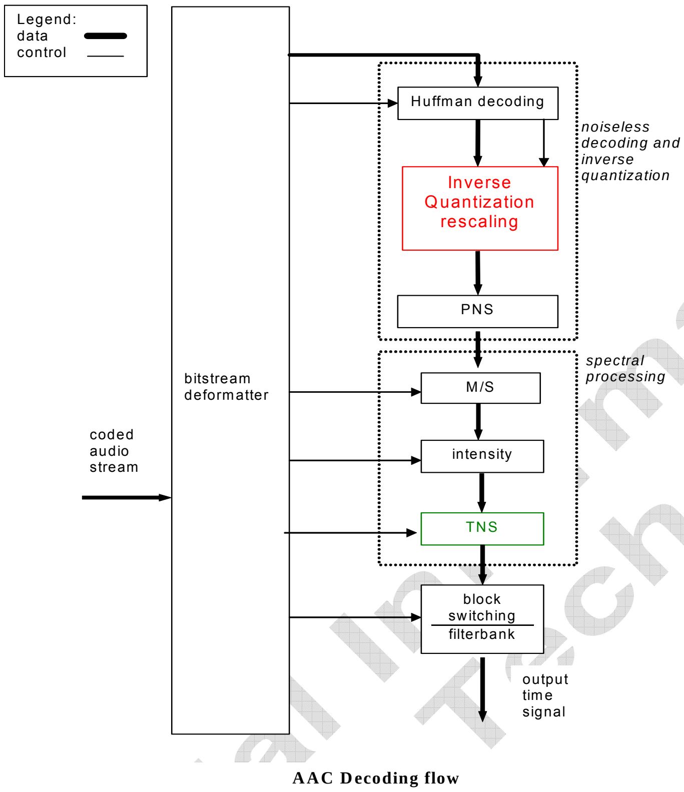


Figure 4-4-2: AAC Decoder Functional Diagram

- ### Main Features
  - Support AAC_LC, AMR, MPEG-1 Level 1, 2 and MP3 decoding.
  - Hardware accelerator for MPEG-1 and MPEG-4 video decoding
  - Hardware accelerator for MP3 and AAC-LC decoding
  - Both MS Stereo and Intensity stereo for Joint Stereo decoding

- ### AAC PNS Processing

The PNS (Perceptual Noise Substitution) tool of MPEG-4 AAC is used to implement perceptual noise substitution coding within an ICS (individual_channel_stream)

- ### AAC TNS Processing

TNS or Temporal Noise Shaping uses the time-frequency duality principle to combat pre-echo artifacts that can result from transient signals (such as castanets). Basically, as time-domain prediction flattens the frequency domain envelope, frequency-domain prediction will flatten the time-domain envelope, thus alleviating the “spreading” effect in time-domain due to lossy frequency-domain processing.

- ### Synthesis Filterbank (MP3 only)

The synthesis filterbank converts the subband time-samples to the original PCM sequence.

- ### DMA interface for MP3 decoder

Input data are send to Re-quantization module via DMA interface, SW is responsible for setting up for DMA operation. The DMA interface includes two parts. One is used to accept data steam from software. The other is used to transform PCM data to system SDRAM

4.6 CPU

The MP612 embeds a 32-bit RISC CPU. It is capable of executing MIPS-I Instruction Set. But the unaligned load and store instructions, LWL, LWR, SWL and SWR, are not supported. Other than that, it is compliant to the MIPS-I software programming model, including execution and exception model. It also includes 2KB of Instruction and Data Cache respectively to enhance the performance further.

It also includes a 32-bit Multiply/Divide/Accumulate (MAC-DIV) module. 32-bit x 32-bit multiplication and division can be accomplished with latency of 5 and 35 CPU clocks respectively. The MAC instruction, with 16-bit x 16-bit multiply being added to 32-bit accumulator can be accomplished with latency of 3 CPU clocks. Thus, it can be utilized to boost the software performance for intensive signal processing tasks, like MP3/MPEG-4 decoding, etc.

The operating frequency of CPU can be dynamically adjusted. Lower frequency setting can be used to save the power consumption. SLEEP instruction can also be used to put the CPU into sleep mode, which will reduce the power consumption of CPU to the minimum level. Any enabled interrupt can then be used to wake up the CPU to normal operation.

A wide range of 3rd party tools can be accessed for program development. These tools include GNU C Compiler and the Green Hills C Compiler. Variety of RTOS, like Wind River, Nucleus Plus, and etc. can also be supported. Complete firmware, driver and debug utilities for software development will also be supported.

4.7 Memory Mapping

The CPU supports 4GB of virtual addressing. It is divided into 4 segments as the following table. The 4GB of virtual address space are mapped to 512MB of physical space by ignoring the 3 MSBs.

Virtual Address Space	Description	Mapped Physical Address
0xE000_0000 ~ 0xFFFF_FFFF	KSEG2. 1GB. Addressable in Kernel mode. Cached	0x0000_0000 ~ 0x1FFF_FFFF
0xC000_0000 ~ 0xDFFF_FFFF		0x0000_0000 ~ 0x1FFF_FFFF
0xA000_0000 ~ 0xBFFF_FFFF	KSEG1. 512MB. Addressable in Kernel mode. Uncached	0x0000_0000 ~ 0x1FFF_FFFF
0x8000_0000 ~ 0x9FFF_FFFF	KSEG0. 512MB. Addressable in Kernel mode. Cached	0x0000_0000 ~ 0x1FFF_FFFF
0x6000_0000 ~ 0x7FFF_FFFF	KUSEG. 2GB. Addressable in Kernel or User mode. Cached	0x0000_0000 ~ 0x1FFF_FFFF
0x4000_0000 ~ 0x5FFF_FFFF		0x0000_0000 ~ 0x1FFF_FFFF
0x2000_0000 ~ 0x3FFF_FFFF		0x0000_0000 ~ 0x1FFF_FFFF
0x0000_0000 ~ 0x1FFF_FFFF		0x0000_0000 ~ 0x1FFF_FFFF

The 512MB of physical space is further mapped as follows:

Physical Space	Range	Description
0x1C00_0000 ~ 0x1FFF_FFFF	448 ~ 512MB	Code Flash/ROM
0x1400_0000 ~ 0x1BFF_FFFF	320 ~ 448MB	Reserved
0x1000_0000 ~ 0x13FF_FFFF	256 ~ 320MB	Peripheral
0x0800_0000 ~ 0x0FFF_FFFF	128 ~ 256MB	Register File
0x0000_0000 ~ 0x07FF_FFFF	0 ~ 128MB	DRAM

The following table also summarizes the register file mapping.

Address Offset	Contents
224 KB ~	Reserved
208 ~ 224 KB	MPA Registers
192 ~ 208 KB	MPV Registers
144 ~ 192 KB	Reserved
136 ~ 144 KB	CDU Registers
128 ~ 136 KB	Memory Card Control Registers
120 ~ 128 KB	IDU/OSD Registers
112 ~ 120 KB	GPIO Registers
104 ~ 112 KB	RTC Registers
96 ~ 104 KB	USB Registers
88 ~ 96 KB	Reserved
80 ~ 88 KB	AIU Registers
72 ~ 80 KB	UART Registers
64 ~ 72 KB	SIO Registers
56 ~ 64 KB	Timer Registers
48 ~ 56 KB	DMA Control Registers
40 ~ 48 KB	Clock Generator/Interrupt Handler Control Registers
32 ~ 40 KB	BIU/System Control Registers
0 ~ 32 KB	IPU Registers and SRAM

The software program can be relocated arbitrarily and flexibly to any part of the physical memory, like SDRAM, internal SRAM or even memory card, not just limited to the Code Flash/ROM. By shadowing the code to DRAM or internal SRAM, it will increase the performance, and also reduce power for the latter case.

4.8 DMA Controller

The DMA Controller manages all the SDRAM access requests from internal module or external peripheral. It arbitrates among the requests with the pre-defined priority. Through each channel, data is transferred from DRAM to device, or vice versa. The DMA channels that require real-time and high data bandwidth are assigned with high priorities. Memory to memory DMA can also be supported in certain configuration. It can also be enabled to take advantage of the bank-interleave feature of SDRAM to increase the memory bandwidth. To save the SDRAM power consumption, several levels of power down mode can also be employed. A quite flexible addressing scheme is supported to maximize SDRAM space usage. It supports a wide range of SDRAM types, from 16Mb to 256Mb, x16 or x 32-bit bus. It also supports DDR SDRAM, from 16Mb to 256Mb, x16-bit bus. It allows the customer to make the best choice for their application. The

SDRAM timing is also programmable to be fit to variety of SDRAM performance.
Besides SDRAM, it also takes care of the ROM/Code Flash and external peripheral accesses. The DMA Controller arbitrates among the 3 types of requests automatically. A Memory Bus Interface Unit is included to handle the interfaces with these 3 kinds of devices. It supports multiple configuration and programmable timing for maximum compatibility. Up to 64MB of Code Flash/ROM can be supported. Not only for firmware code, it can also be used for image storage if desired. It also supports quite flexible peripheral interface. Peripheral with DMA capability can also be supported. It can thus expand the system functionality.

4.9 Memory Card Controller

The Memory Card Controller is responsible of controlling the data accesses with external image storage media. It supports Compact Flash, Smart Media, Secure Digital Memory Card (SD), Multi Media Card (MMC), XD , Memory Stick and IBM Micro Drive. Coexist with the memory card, one on-board NAND Flash can also be supported.
For Compact Flash interface, either Memory or True IDE mode can be supported. By default, it supports 8-bit data bus width. It can also be configured to select 16-bit interface. Hardware ECC and CRC are implemented for Smart Media and MMC/SD support respectively.
Data transfer with Compact Flash or Smart Media can be accomplished through DMA or CPU. For MMC or SD, however, it allows only through DMA. The bus interface timing is programmable for maximum compatibility.

4.10 USB Device Controller

The built-in USB Device Controller support several endpoints for communication with USB Host.

- Control Read/Write transfer
- Interrupt Transfer
- Downstream Bulk Transfer from Host
- Upstream Bulk Transfer to Host
- Upstream Video Isochronous /Bulk Transfer to Host
- Upstream Audio Isochronous /Bulk Transfer to Host

The upstream bulk or isochronous transfer is accomplished with DMA operation.

4.11 Audio Interface Unit (AIU)

The Audio Interface Unit (AIU) handles data communication between MP612 and external audio CODEC. Both audio playback and recording are supported. It implements a quite flexible interface for standard AC97 or non-standard audio CODEC, for example OKI-MSM7732, TI-TLV320AC36/37, Philips-UDA1342TS, AKM-AK4516A, etc. It supports both Master and Slave clocking. It also supports variety of audio data format, 8 or 16-bit, mono or stereo, and sampling rate for both recording and playback.

4.12 GPIO

The MP612 provides 32 dedicated GPIO pins. There are also additional 103 pins can be configured as GPIO if the associated function is not used. Most of the pins can be programmed for alternative functions by setting corresponding configuration register. Those dedicated GPIO pins can be enabled to generate interrupt with level or edge trigger. The polarity of level or edge interrupt is programmable to fit variety of application requirement.

The GPIO input pins with interrupt capability, when enabled, can also be used to wake up the chip from deep power down mode.

Other than GPIO, the MP612 also supports 2 PWM outputs to control device like motor, buzzer, etc. With the parallel interface controlled by the DMA Controller, the MP612 is able to support almost all kinds of peripherals.

5 Operation Modes

To fulfill the functions that a digital photo frame requires, the MP612 can be used in, but not limited to the following operation modes:

- **Playback/OS command mode**
  - The stored images/video/audio files can be played back to LCD panel, speaker or TV output port in this mode
  - Executable software can be run at this mode for any application
  - Manage the available modules of DPF
- **USB upload mode**
  - The stored images/video/audio can be uploaded to PC for further application
- **USB download mode**
  - DPF firmware can be updated by downloading the new firmware code and activated automatically
  - Executable codes for 32-bit RISC can be downloaded from PC and run at embedded OS environment
  - Multimedia or MP3 files can be playback by using the existing hardware inside the DPF

6 Ball Assignment Diagram (Top View)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	MD[24]	MD[27]	MD[29]	MROMCSX	TESTMD[1]	RESETX	GP30	GP24	GP22	GP18	GP16	GP3	GP2	FXDWPX	FSDWPX	FSMWPX
B	MD[19]	MD[25]	MD[31]	MDQM[3]	MD[23]	SCANEN	TESTMD[0]	GP31	GP28	GP20	URX	GP6	GP0	FXDCDX	FXDPWREN	FSMCDX
C	CLKIN	CLKOUT	MD[17]	MD[20]	MD[21]	GP29	RSTOUTX	GP23	NAFBOOT	GP26	UTX	GP4	ADIN	AMCLK	FATDMACKX	FSMD3
D	MRASX	MDQM[0]	MWEX	MSDCK	MCASX	MD[28]	GP21	MDQM[2]	GP25	GP17	GP5	AFSYNC	ACLK	FMSINS	FSDCLK	FSMD1
E	MD[18]	MD[1]	MSDCKX	MSDCKE	MSDCSX	MD[22]	MD[26]	MD[30]	GP27	GP19	GP1	ADOUT	FSDCDX	FXDCEX	FSMWEX	FWAITX
F	MD[0]	MD[16]	MD[3]	TESTMD[2]	APGND2	APGNDP1	APVPP1	GND	GND	GND	GP7	FMSCLK	FATDMARQ	FATCS0X	FSCE0X	FRDY
G	MD[2]	MD[4]	MSDVREF	MD[7]	MGNDP1/2/3	MGNDP1/2/3	APVDD1	APVDD2	GND	GND	GND	FSMALE	FSMD2	FSMD0	FREGX	FIORDX
H	MSDLDQS	MD[6]	MD[8]	MSDUDQS	MGNDP1/2/3	MVPP	VDD	APGND1	GND	GND	GND	FA[10]	FIOWRX	FRESET	FCD2X	FWEX
J	MD[5]	MD[12]	MA[0]	MD[15]	MVPP	MVPP	VDD	VPP	UHGNDP	UFGNDP	GND	FA[2]	FA[6]	FA[1]	FCE2X	FOEX
K	MDQM[1]	MD[10]	MA[4]	MA[8]	AVVDD1	VDD	VDD	VPP	UHGND	UFVPP	GND	FD[8]	FD[12]	FA[0]	FA[9]	FCD1X
L	MD[9]	MD[11]	MA[3]	MA[10]	AVGND1	AVVPP1	VPP	VPP	UHVDD	RVPP	VPP	FD[4]	FD[0]	FD[6]	FA[8]	FCE1X
M	MD[13]	MA[1]	MA[6]	MBA[1]	AVGND3	AVVPP2	AVGND4	VPP	VDD	RVDD	UHVPP	FD[10]	USBFDP	FD[9]	FD[11]	FA[7]
N	MD[14]	MA[2]	MA[9]	VIREF	VD0	VD6	VHSYNC	VCCIR_C3	GP9	RXOUT	UREFEXT	USBHDP	URDATAP	FD[7]	FA[4]	FA[5]
P	MA[5]	MA[7]	MA[11]	VCOMPA	AVIGOUT	VD3	VD7	VVSYNC	VCCIR_C7	UHSXTALI	GP14	RXIN	URDATAN	FD[3]	FD[14]	FA[3]
R	MBA[0]	MA[12]	VD1	VD5	VDCK	VCCIR_C0	VCCIR_C2	VCCIR_C5	GP10	GP11	GP13	UHSXTALO	USBHDM	FD[1]	FD[5]	FD[15]
T	AVIBOUT	AVIROUT	VD2	VD4	VDVALID	VCCIR_C1	VCCIR_C4	VCCIR_C6	GP8	GP12	GP15	RALARM	USBFDM	UHVBUS	FD[2]	FD[13]

7 Pin Configuration

Pin Definition

1. Memory Interface (62)

Pin Name	*Default*					*Alt. Func. 1*	*Alt. Func. 2*	*Type*
	Function	*SDRAM*	*PROM/NOR Flash*	*Non-A/D-Muxed Peripheral*	*A/D-Muxed Peripheral*			
MD[7:0]	MD[7:0]	MD[7:0]	MD[7:0]	MD[7:0]	AD[7:0]			IO
MD[14:8]	MD[14:8]/ MA[25:19]	MD[14:8]	MA[25:19]	MA[25:19]				IO
MD[15]	MD[15]	MD[15]						IO
MD[16]	MD[16]	MD[16]	MD[8]	MD[8]		MGPIO[0]		IO
MD[17]	MD[17]	MD[17]	MD[9]	MD[9]		MGPIO[1]		IO
MD[18]	MD[18]	MD[18]	MD[10]	MD[10]		MGPIO[2]		IO
MD[19]	MD[19]	MD[19]	MD[11]	MD[11]		MGPIO[3]		IO
MD[20]	MD[20]	MD[20]	MD[12]	MD[12]		MGPIO[4]		IO
MD[21]	MD[21]	MD[21]	MD[13]	MD[13]		MGPIO[5]		IO
MD[22]	MD[22]	MD[22]	MD[14]	MD[14]		MGPIO[6]		IO
MD[23]	MD[23]	MD[23]	MD[15]	MD[15]		MGPIO[7]		IO
MD[31:24]	MD[31:24]/ MDMAD[7:0]/ CFD[15:8]	MD[31:24]				MGPIO[15:8]		IO
MA[7:0]	MA[7:0]	MA[7:0]	MA[7:0]	MA[7:0]	A[15:8]	-		IO
MA[12:8]	MA[12:8]	MA[12:8]	MA[12:8]	MA[12:8]				IO
MBA[0]	BA[0]/MA[13]	BA[0]	MA[13]	MA[13]				TO
MBA[1]	BA[1]/MA[14]	BA[1]	MA[14]	MA[14]				TO
MRASx	RAS#	RAS#						TO
MWEx	MWE#	MWE#	MWE#	MWE#	MWE#			TO
MCASx	CAS#/MOE#	CAS#	MOE#	MOE#	MOE#			TO
MDQM[0]/ LDQM	DQM[0]/ MA[15]	DQM[0]	MA[15]	MA[15]				TO
MDQM[1]/ UDQM	DQM[1]/ MA[16]	DQM[1]	MA[16]	MA[16]				TO
MDQM[2]	DQM[2]/ MA[17]	DQM[2]	MA[17]	MA[17]				TO
MDQM[3]	DQM[3]/ MA[18]	DQM[3]	MA[18]	MA[18]				TO
MSDCK	SDCLK	SDCLK						IO
MSDCKx	SDCLKx	SDCLKx						IO
MSDCKE	SDCKE	SDCKE						TO
MSDCSx	SDRAMCS#	SDRAMCS#						TO
MSDVREF								A
MSDLDQS	LDQS	LDQS						IO
MSDUDQS	UDQS	UDQS						IO
MROMCSx	MROMCS#		MROMCS#					TO

Pin Name		Pin Number	I/O	Definition
MD[7:0]	MD[7:0]	G4, H2, J1, G2,	I/O	DRAM Data Bus bit 7~0
		F3, G1, E2, F1	I/O	PROM/NOR Flash Data Bus bit 7~0

			I/O	Non-A/D-Muxed Peripheral Data Bus bit 7~0
			I/O	A/D-Muxed Peripheral Data Bus bit 7~0
MD[14:8]	MD[14:8]	N1,M1, J2, L2, K2, L1, H3	I/O	DRAM Data Bus bit 14~8
	MA[25:19]		I/O	PROM/NOR Flash Address bit 25~19
	MA[25:19]		I/O	Non-A/D-Muxed Peripheral Address bit 25~19
MD[15]	MD[15]	J4	I/O	DRAM Data Bus bit 15
MD[16]	MD[16]	F2	I/O	DRAM Data Bus bit 16
	MD[8]		I/O	PROM/NOR Flash Data Bus bit 8
	MD[8]		I/O	Non-A/D-Muxed Peripheral Data Bus bit 8
	MGPIO[0]		I/O	General Purpose I/O pin 0 shared with memory pin
MD[17]	MD[17]	C3	I/O	DRAM Data Bus bit 17
	MD[9]		I/O	PROM/NOR Flash Data Bus bit 9
	MD[9]		I/O	Non-A/D-Muxed Peripheral Data Bus bit 9
	MGPIO[1]		I/O	General Purpose I/O pin 1 shared with memory pin
MD[18]	MD[18]	E1	I/O	DRAM Data Bus bit 18
	MD[10]		I/O	PROM/NOR Flash Data Bus bit 10
	MD[10]		I/O	Non-A/D-Muxed Peripheral Data Bus bit 10
	MGPIO[2]		I/O	General Purpose I/O pin 2 shared with memory pin
MD[19]	MD[19]	B1	I/O	DRAM Data Bus bit 19
	MD[11]		I/O	PROM/NOR Flash Data Bus bit 11
	MD[11]		I/O	Non-A/D-Muxed Peripheral Data Bus bit 11
	MGPIO[3]		I/O	General Purpose I/O pin 3 shared with memory pin
MD[20]	MD[20]	C4	I/O	DRAM Data Bus bit 20
	MD[12]		I/O	PROM/NOR Flash Data Bus bit 12
	MD[12]		I/O	Non-A/D-Muxed Peripheral Data Bus bit 12
	MGPIO[4]		I/O	General Purpose I/O pin 4 shared with memory pin
MD[21]	MD[21]	C5	I/O	DRAM Data Bus bit 21
	MD[13]		I/O	PROM/NOR Flash Data Bus bit 13
	MD[13]		I/O	Non-A/D-Muxed Peripheral Data Bus bit 13
	MGPIO[5]		I/O	General Purpose I/O pin 5 shared with memory pin
MD[22]	MD[22]	E6	I/O	DRAM Data Bus bit 22
	MD[14]		I/O	PROM/NOR Flash Data Bus bit 14
	MD[14]		I/O	Non-A/D-Muxed Peripheral Data Bus bit 14

	MGPIO[6]		I/O	General Purpose I/O pin 6 shared with memory pin
MD[23]	MD[23]	B5	I/O	DRAM Data Bus bit 23
	MD[15]		I/O	PROM/NOR Flash Data Bus bit 15
	MD[15]		I/O	Non-A/D-Muxed Peripheral Data Bus bit 15
	MGPIO[7]		I/O	General Purpose I/O pin 7 shared with memory pin
MD[31:24]	MD[31:24]	B3, E8, A3, D6,	I/O	DRAM Data Bus bit 31~24
	MGPIO[15:8]	A2, E7, B2, A1	I/O	General Purpose I/O pin 15~8 shared with memory pin
MA[7:0]	MA[7:0]	P2, M3, P1,	I/O	DRAM Address Bus bit 7~0
		K3, L3, N2, M2,	I/O	PROM/NOR Flash Address Bus bit 7~0
		J3	I/O	Non-A/D-Muxed Address Bus bit 7~0
	AD[15:8]		I/O	A/D-Muxed Peripheral Adress 15~8
MA[12:8]	MA[12:8]	R2, P3, L4, N3,	I/O	DRAM Address Bus bit 12~8
		K4	I/O	PROM/NOR Flash Address Bus bit 12~8
			I/O	Non-A/D-Muxed Address Bus bit 12~8
MBA[0]	BA[0]	R1	T/O	DRAM Bank Selection 0
	MA[13]		T/O	PROM/NOR Flash Address Bus bit 13
	MA[13]		T/O	Non-A/D-Muxed Peripheral Address Bus bit 13
MBA[1]	BA[1]	M4	T/O	DRAM Bank Selection 1
	MA[14]		T/O	PROM/NOR Flash Address Bus bit 14
	MA[14]		T/O	Non-A/D-Muxed Peripheral Address Bus bit 14
MRASx	RAS#	D1	T/O	DRAM Row Address Strobe (Active low)
MWEx	MWE#	D3	T/O	DRAM Write Enable (Active low)
			T/O	PROM/NOR Flash Write Enable (Active low)
			T/O	Non-A/D-Muxed Peripheral Write Enable (Active low)
			T/O	A/D-Muxed Peripheral Write Enable (Active low)
MCASx	CAS#	D5	T/O	DRAM Column Address Strobe (Active low)
	MOE#		T/O	PROM/NOR Flash Output Enable (Active low)
			T/O	Non-A/D-Muxed Peripheral Output Enable (Active low)
			T/O	A/D-Muxed Peripheral Output Enable (Active low)
MDQM[0]	DQM[0]	D2	T/O	DRAM Data Input/Output Mask 0
	MA[15]		T/O	PROM/NOR Flash Address Bus bit 15
	MA[15]		T/O	Non-A/D-Muxed Peripheral Address Bus bit 15
MDQM[1]	DQM[1]	K1	T/O	DRAM Data Input/Output Mask 1

	MA[16]		T/O	PROM/NOR Flash Address Bus bit 16
	MA[16]		T/O	Non-A/D-Muxed Peripheral Address Bus bit 16
MDQM[2]	DQM[2]	D8	T/O	DRAM Data Input/Output Mask 2
	MA[17]		T/O	PROM/NOR Flash Address Bus bit 17
	MA[17]		T/O	Non-A/D-Muxed Peripheral Address Bus bit 17
MDQM[3]	DQM[3]	B4	T/O	DRAM Data Input/Output Mask 2
	MA[18]		T/O	PROM/NOR Flash Address Bus bit 17
	MA[18]		T/O	Non-A/D-Muxed Peripheral Address Bus bit 17
MSDCK	SDCLK	D4	I/O	(1) SDCLK: Output Clock to SDRAM (2) SDCLK&SDCLK#: Output Clock to DDR SDRAM SDCLK and SDCLK# are differential clock Outputs.
MSDCKx	SDCLK#	E3	I/O	
MSDCKE	SDCKE	E4	T/O	Output Clock to DRAM Enable (Active Low)
MSDCSx	SDRAMCS#	E5	T/O	DRAM Chip Select
MSDVREF	MSDVREF	G3	Analog	DRAM Reference Voltage
MSDLDQS	LDQS	H1	I/O	Data Strobe : Output with read data, input with write data.
MSDUDQS	UDQS	H4	I/O	Edge-aligned with read data, centered in write data. Used to capture write data.
MROMCSx	MROMCS#	A4	T/O	PROM/NOR Flash Chip Select

2. Memory Card and Hard-Disk Interface (CF/SM/SD/MMC/MS Pro/xD) (60)

Pin Name	*Default*	*Alt. Func. 1*							*Alt. Func. 2*	*Type*
		Compact Flash		*Smart Media (NAND Flash)*	*MMC/SD*	*Memory Stick PRO*	*XD*	*ATA-66*		
		Memory	*True IDE*							
FD[0]	FGPIO[0]	FD[0]	FD[0]			SDIO[0]	XD[0]	DD[0]		IO
FD[3:1]	FGPIO[3:1]	FD[3:1]	FD[3:1]			SDIO[3:1]	XD[3:1]	DD[3:1]		IO
FD[4]	FGPIO[4]	FD[4]	FD[4]	SD[4]			XD[4]	DD[4]		IO
FD[7:5]	FGPIO[7:5]	FD[7:5]	FD[7:5]	SD[7:5]			XD[7:5]	DD[7:5]		IO
FD[8]	FGPIO[8]	FD[8]	FD[8]		DAT[0]			DD[8]		IO
FD[9]	FGPIO[9]	FD[9]	FD[9]		DAT[1]			DD[9]		IO
FD[10]	FGPIO[10]	FD[10]	FD[10]		DAT[2]			DD[10]		IO
FD[11]	FGPIO[11]	FD[11]	FD[11]		DAT[3]			DD[11]		IO
FD[12]	FGPIO[12]	FD[12]	FD[12]		DAT[4]			DD[12]		IO
FD[13]	FGPIO[13]	FD[13]	FD[13]		DAT[5]			DD[13]		IO
FD[14]	FGPIO[14]	FD[14]	FD[14]		DAT[6]			DD[14]		IO
FD[15]	FGPIO[15]	FD[15]	FD[15]		DAT[7]			DD[15]		IO
FA[0]	FGPIO[16]	FA[0]	FA[0]				XCLE	DA[0]		IO
FA[1]	FGPIO[17]	FA[1]	FA[1]				XALE	DA[1]		IO
FA[2]	FGPIO[18]	FA[2]	FA[2]		CMD			DA[2]		IO
FA[3]	FGPIO[19]	FA[3]	0							IO
FA[4]	FGPIO[20]	FA[4]	0			BS				IO

FA[5]	FGPIO[21]	FA[5]	0				XRB			IO
FA[6]	FGPIO[22]	FA[6]	0	SRB						IO
FA[7]	FGPIO[23]	FA[7]	0	SCE1#						IO
FA[8]	FGPIO[24]	FA[8]	0	SCLE						I/O
FA[9]	FGPIO[25]	FA[9]	0							I/O
FA[10]	FGPIO[26]	FA[10]	0							IO
FCE1x	FGPIO[27]	FCE1#	FCS0#							IO
FCE2x	FGPIO[28]	FCE2#	FCS1#							IO
FCD1x	FGPIO[29]	FCD1#	FCD1#	-						IO
FCD2x	FGPIO[30]	FCD2#	FCD2#		-					IO
FOEx	FGPIO[31]	FOE#	ATASEL# (=0)		-		XRE#			IO
FWEx	FGPIO[32]	FWE#	1		-		XWE#			IO
FIORDx	FGPIO[33]		FIORD#	SRE#				DIOR#		IO
FIOWRx	FGPIO[34]		FIOWR#					DIOW#		IO
FRDY	FGPIO[35]	FRDY/BUSY#	FINTRQ		-					IO
FRESET	FGPIO[36]	FRESET	FRESET#	-	-					IO
FWAITx	FGPIO[37]	FWAIT#	(FIORDY)		-			IORDY		IO
FREGx	FGPIO[38]	FREG#	1	-	-					IO
FSCE0x	FGPIO[39]			SCE0#						IO
FSMD[0]	FGPIO[40]			SD[0]						IO
FSMD[1]	FGPIO[41]			SD[1]						IO
FSMD[2]	FGPIO[42]			SD[2]						IO
FSMD[3]	FGPIO[43]			SD[3]						IO
FSMALE	FGPIO[44]			ALE						IO
FSMWEx	FGPIO[45]			WE#						IO
FSDCLK	FGPIO[46]				CLK					IO
FMSCLK	FGPIO[47]					SCLK				IO
FATDMARQ	FGPIO[48]							DMARQ		IO
FATDMACKx	FGPIO[49]							DMACK#		IO
FATCS0x	FGPIO[50]							CS0x		IO
FXDCEx	FGPIO[51]						XCE#			IO
FXDPWREN	FGPIO[52]						XVCCtrl			IO
FSMCDx	FGPIO[53]			CDx						IO
FSMWPx	FGPIO[54]			WP#						IO
FSDCDx	FGPIO[55]				CD#					IO
FSDWPx	FGPIO[56]				WP#					IO
FMSINS	FGPIO[57]					MS_INS				IO
FXDWPx	FGPIO[58]						XWP#			IO
FXDCDx	FGPIO[59]						XCDx			IO

Note: FGPIO[59:52] has interrupt capability

Pin Name		Pin Number	I/O	Definition
FD[0]	FD[0]	L13	I/O	COMPACT FLASH(MEMORY) Data bus 0
	FD[0]		I/O	COMPACT FLASH (TRUE IDE) Data bus 0
	SDIO[0]		I/O	MEMORY STICK PRO Data I/O 0
	XD[0]		I/O	XD Data bus 0
	DD[0]		I/O	ATA-66 Data Bus 0
	FGPIO[0]		I/O	General purpose I/O 0 shared with memory card

FD[3:1]	FD[3:1]	P14, T15, R14	I/O	COMPACT FLASH(MEMORY) Data bus 3~1
	FD[3:1]		I/O	COMPACT FLASH (TRUE IDE) Data bus 3~1
	SDIO[3:1]		I/O	MEMORY STICK PRO Data I/O 3~1
	XD[3:1]		I/O	XD Data bus 3~1
	DD[3:1]		I/O	ATA-66 Data Bus 3~1
	FGPIO[3:1]		I/O	General purpose I/O 3~1 shared with memory card
FD[4]	FD[4]	L12	I/O	COMPACT FLASH(MEMORY) Data bus4
	FD[4]		I/O	COMPACT FLASH (TRUE IDE) Data bus 4
	SD[4]		I/O	SMART MEDIA(NAND FLASH) Data bus 4
	XD[4]		I/O	XD Data bus 4
	DD[4]		I/O	ATA-66 Data Bus4
	FGPIO[4]		I/O	General purpose I/O 4 shared with memory card
FD[7:5]	FD[7:5]	N14, L14, R15	I/O	COMPACT FLASH(MEMORY) Data bus 7~5
	FD[7:5]		I/O	COMPACT FLASH (TRUE IDE) Data bus 7~5
	SD[7:5]		I/O	SMART MEDIA(NAND FLASH) Data bus 7~5
	XD[7:5]		I/O	XD Data bus 7~5
	DD[7:5]		I/O	ATA-66 Data Bus 7~5
	FGPIO[7:5]		I/O	General purpose I/O 7~5 shared with memory card
FD[8]	FD[8]	K12	I/O	COMPACT FLASH(MEMORY) Data bus 8
	FD[8]		I/O	COMPACT FLASH (TRUE IDE) Data bus 8
	DAT[0]		I/O	MMC Data Bus 0
	DAT[0]		I/O	SD Data Bus 0
	DD[8]		I/O	ATA-66 Data Bus 8
	FGPIO[8]		I/O	General purpose I/O 8 shared with memory card
FD[9]	FD[9]	M14	I/O	COMPACT FLASH(MEMORY) Data bus 9
	FD[9]		I/O	COMPACT FLASH (TRUE IDE) Data bus 9
	DAT[1]		I/O	MMC Data Bus 1
	DAT[1]		I/O	SD Data Bus 1
	DD[9]		I/O	ATA-66 Data Bus 9
	FGPIO[9]		I/O	General purpose I/O 9 shared with memory card
FD[10]	FD[10]	M12	I/O	COMPACT FLASH(MEMORY) Data bus 10
	FD[10]		I/O	COMPACT FLASH (TRUE IDE) Data bus 10
	DAT[2]		I/O	MMC Data Bus 2

	DAT[2]		I/O	SD Data Bus 2
	DD[10]		I/O	ATA-66 Data Bus 10
	FGPIO[10]		I/O	General purpose I/O 10 shared with memory card
FD[11]	FD[11]	M15	I/O	COMPACT FLASH(MEMORY) Data bus 11
	FD[11]		I/O	COMPACT FLASH (TRUE IDE) Data bus 11
	DAT[3]		I/O	MMC Data Bus 3
	DAT[3]		I/O	SD Data Bus 3
	DD[11]		I/O	ATA-66 Data Bus 11
	FGPIO[11]		I/O	General purpose I/O 11 shared with memory card
FD[12]	FD[12]	K13	I/O	COMPACT FLASH(MEMORY) Data bus 12
	FD[12]		I/O	COMPACT FLASH (TRUE IDE) Data bus 12
	DAT[4]		I/O	MMC Data Bus 4
	DD[12]		I/O	ATA-66 Data Bus 12
	FGPIO[12]		I/O	General purpose I/O 12 shared with memory card
FD[13]	FD[13]	T16	I/O	COMPACT FLASH(MEMORY) Data bus 13
	FD[13]		I/O	COMPACT FLASH (TRUE IDE) Data bus 13
	DAT[5]		I/O	MMC Data Bus 5
	DD[13]		I/O	ATA-66 Data Bus 13
	FGPIO[13]		I/O	General purpose I/O 13 shared with memory card
FD[14]	FD[14]	P15	I/O	COMPACT FLASH(MEMORY) Data bus 14
	FD[14]		I/O	COMPACT FLASH(TRUE IDE) Data Bus 14
	DAT[6]		I/O	MMC Data Bus 6
	DD[14]		I/O	ATA-66 Data Bus 14
	FGPIO[14]		I/O	General purpose I/O 14 shared with memory card
FD[15]	FD[15]	R16	I/O	COMPACT FLASH(MEMORY) Data bus 15
	FD[15]		I/O	COMPACT FLASH (TRUE IDE) Data bus 14
	DAT[7]		I/O	MMC Data Bus 7
	DD[15]		I/O	ATA-66 Data Bus 15
	FGPIO[15]		I/O	General purpose I/O 15 shared with memory card
FA[0]	FA[0]	K14	I/O	COMPACT FLASH(MEMORY) Address 0 FA[0:10] along with the REG# signal are used to select the following: The I/O port address registers within the CompactFlash Storage Card or CF+ Card, the memory mapped port address registers within the CompactFlash Storage Card or CF+ Card, a byte in the card's information structure and its configuration control and

				status registers.
	FA[0]		I/O	COMPACT FLASH (TRUE IDE) Address 0 In True IDE Mode only FA[0] ~ FA[2] are used to select the one of eight registers in the Task File.
	XCLE		I/O	XD Clock Enable
	DA[0]		I/O	ATA-66 Address 0
	FGPIO[16]		I/O	General purpose I/O 16 shared with memory card
FA[1]	FA[1]	J14	I/O	COMPACT FLASH(MEMORY) Address 1
	FA[1]		I/O	COMPACT FLASH (TRUE IDE) Address 1 In True IDE Mode only FA[0] ~ FA[2] are used to select the one of eight registers in the Task File.
	XALE		I/O	XD Address Latch Enable
	DA[1]		I/O	ATA-66 Address 1
	FGPIO[17]		I/O	General purpose I/O 17 shared with memory card
FA[2]	FA[2]	J12	I/O	COMPACT FLASH(MEMORY) Address 2
	FA[2]		I/O	COMPACT FLASH (TRUE IDE) Address 2 In True IDE Mode only FA[0] ~ FA[2] are used to select the one of eight registers in the Task File.
	CMD		I/O	MMC Command
	CMD		I/O	SD Command
	DA[2]		I/O	ATA-66 Address 2
	FGPIO[18]		I/O	General purpose I/O 18 shared with memory card
FA[3]	FA[3]	P16	I/O	COMPACT FLASH(MEMORY) Address 3
	FGPIO[19]		I/O	General purpose I/O 19 shared with memory card
FA[4]	FA[4]	N15	I/O	COMPACT FLASH(MEMORY) Address 4
	BS		I/O	MEMORY STICK PRO Bus State
	FGPIO[20]		I/O	General purpose I/O 20 shared with memory card
FA[5]	FA[5]	N16	I/O	COMPACT FLASH(MEMORY) Address 5
	XRB		I/O	XD Ready & Busy
	FGPIO[21]		I/O	General purpose I/O 21 shared with memory card
FA[6]	FA[6]	J13	I/O	COMPACT FLASH(MEMORY) Address 6
	SRB		I/O	SMART MEDIA(NAND FLASH) Ready & Busy
	FGPIO[22]		I/O	General purpose I/O 22 shared with memory card
FA[7]	FA[7]	M16	I/O	COMPACT FLASH(MEMORY) Address 7

	SCE1#		I/O	SMART MEDIA(NAND FLASH)Chip Enable Strobe(Active low)
	FGPIO[23]		I/O	General purpose I/O 23 shared with memory card
FA[8]	FA[8]	L15	I/O	COMPACT FLASH(MEMORY) Address 8
	SCLE		I/O	SMART MEDIA(NAND FLASH) Command Latch Enable
	FGPIO[24]		I/O	General purpose I/O 24 shared with memory card
FA[9]	FA[9]	K15	I/O	COMPACT FLASH(MEMORY) Address 9
	FGPIO[25]		I/O	General purpose I/O 25 shared with memory card
FA[10]	FA[10]	H12	I/O	COMPACT FLASH(MEMORY) Address 10
	FGPIO[26]		I/O	General purpose I/O 26 shared with memory card
FCE1x	FCE1#	L16	I/O	COMPACT FLASH(MEMORY) Card Enable CE1# accesses the even byte or the Odd byte of the word depending on FA0 and CE2#
	FCS0#		I/O	COMPACT FLASH(TRUE IDE) In the True IDE Mode CS0 is the chip select for the task file registers
	FGPIO[27]		I/O	General purpose I/O 27 shared with memory card
FCE2x	FCE2#	J15	I/O	COMPACT FLASH(MEMORY) Card Enable CE2# always accesses the odd byte of the word.
	FCS1#		I/O	COMPACT FLASH(TRUE IDE) In the True IDE Mode CS2 is used to select the Alternate Status Register and the Device Control Register.
	FGPIO[28]		I/O	General purpose I/O 28 shared with memory card
FCD1x	FCD1#	K16	I/O	COMPACT FLASH(MEMORY) Card Detect 1
	FCD1#		I/O	COMPACT FLASH (TRUE IDE) Card Detect 1
	FGPIO[29]		I/O	General purpose I/O 29 shared with memory card
FCD2x	FCD2#	H15	I/O	COMPACT FLASH(MEMORY) Card Detect 2
	FCD2#		I/O	COMPACT FLASH (TRUE IDE) Card Detect 2
	FGPIO[30]		I/O	General purpose I/O 30 shared with memory card
FOEx	FOE#	J16	I/O	COMPACT FLASH(MEMORY) Output Enable (Active low)
	ATASEL#(=0)		I/O	COMPACT FLASH(TRUE IDE) ATA Select (Active low)
	XRE#		I/O	XD Read Enable Strobe (Active low)
	FGPIO[31]		I/O	General purpose I/O 31 shared with memory card

FWEx	FWE#	H16	I/O	COMPACT FLASH(MEMORY) Write Enable Strobe (Active low)
	1		I/O	COMPACT FLASH(TRUE IDE) Connect to High
	XWE#		I/O	XD write Enable Strobe (Active low)
	FGPIO[32]		I/O	General purpose I/O 32 shared with memory card
FIORDx	FIORD#	G16	I/O	COMPACT FLASH(TRUE IDE) I/O Read Enable Strobe (Active low)
	SRE#		I/O	SMART MEDIA(NAND FLASH) Read Enable Strobe (Active Low)
	DIOR#		I/O	ATA-66 I/O Ready enable strobe (Active low)
	FGPIO[33]		I/O	General purpose I/O 33 shared with memory card
FIOWRx	FIOWR#	H13	I/O	COMPACT FLASH(TRUE IDE) I/O Write Enable (Active Low)
	DIOW#		I/O	ATA-66 I/O Write enable strobe (Active low)
	FGPIO[34]		I/O	General purpose I/O 34 shared with memory card
FRDY	FRDY/BUSY#	F16	I/O	COMPACT FLASH(MEMORY) Ready & Busy
	FINTRQ		I/O	COMPACT FLASH(TRUE IDE) Interrupt Request
	FGPIO[35]		I/O	General purpose I/O 35 shared with memory card
FRESET	FRESET	H14	I/O	COMPACT FLASH(MEMORY) Reset
	FRESET#		I/O	COMPACT FLASH(TRUE IDE) Reset (Active low)
	FGPIO[36]		I/O	General purpose I/O 36 shared with memory card
FWAITx	FWAIT#	E16	I/O	COMPACT FLASH(MEMORY) Wait Enable (Active Low)
	(FIORDY)		I/O	COMPACT FLASH(TRUE IDE) I/O Ready
	IORDY		I/O	ATA-66 I/O Ready enable strobe (Active low)
	FGPIO[37]		I/O	General purpose I/O 37 shared with memory card
FREGx	FREG#	G15	I/O	COMPACT FLASH(MEMORY) Register (Active low)
	1		I/O	COMPACT FLASH(TRUE IDE) Connect to High
	FGPIO[38]		I/O	General purpose I/O 38 shared with memory card
FSCE0x	SCE0#	F15	I/O	SMART MEDIA(NAND FLASH) Chip Enable Strobe (Active Low)
	FGPIO[39]		I/O	General purpose I/O 39 shared with memory card
FSMD0	SD[0]	G14	I/O	SMART MEDIA(NAND FLASH) Data bus 0
	FGPIO[40]		I/O	General purpose I/O 40 shared with memory card
FSMD1	SD[1]	D16	I/O	SMART MEDIA(NAND FLASH) Data bus 1

	FGPIO[41]		I/O	General purpose I/O 41 shared with memory card
FSMD2	SD[2]	G13	I/O	SMART MEDIA(NAND FLASH) Data bus 2
	FGPIO[42]		I/O	General purpose I/O 42 shared with memory card
FSMD3	SD[3]	C16	I/O	SMART MEDIA(NAND FLASH) Data bus 3
	FGPIO[43]		I/O	General purpose I/O 43 shared with memory card
FSMALE	ALE	G12	I/O	SMART MEDIA(NAND FLASH) Address Latch Enable
	FGPIO[44]		I/O	General purpose I/O 44 shared with memory card
FSMWEx	WE#	E15	I/O	SMART MEDIA(NAND FLASH) Write Enable Strobe (Active low)
	FGPIO[45]		I/O	General purpose I/O 45 shared with memory card
FSDCLK	CLK	D15	I/O	MMC Clock
	CLK		I/O	SD Clock
	FGPIO[46]		I/O	General purpose I/O 46 shared with memory card
FMSCLK	SCLK	F12	I/O	MEMORY STICK PRO Serial Clock
	FGPIO[47]		I/O	General purpose I/O 47 shared with memory card
FATDMARQ	DMARQ	F13	I/O	ATA-66 DMA Request
	FGPIO[48]		I/O	General purpose I/O 48 shared with memory card
FATDMACKx	DMACK#	C15	I/O	ATA-66 Acknowledge (Active Low)
	FGPIO[49]		I/O	General purpose I/O 49 shared with memory card
FATCS0x	CS0#	F14	I/O	ATA-66 Chip Select 0 (Active Low)
	FGPIO[50]		I/O	General purpose I/O 50 shared with memory card
FXDCEx	XCE#	E14	I/O	XD Chip Enable Strobe (Active low)
	FGPIO[51]		I/O	General purpose I/O 51 shared with memory card
FXDPWREN	XVCCtrl	B15	I/O	XD Power Enable
	FGPIO[52]		I/O	General purpose I/O 52 shared with memory card
FSMCDx	CD#	B16	I/O	SMART MEDIA(NAND FLASH) Card Detect (Active low)
	FGPIO[53]		I/O	General purpose I/O 53 shared with memory card
FSMWPx	WP#	A16	I/O	SMART MEDIA(NAND FLASH) Write Protect(Active low)
	FGPIO[54]		I/O	General purpose I/O 54 shared with memory card
FSDCDx	CD#	E13	I/O	MMC Card Detect(Active low)
	CD#		I/O	SD Card Detect(Active low)
	FGPIO[55]		I/O	General purpose I/O 55 shared with memory card
FSDWPx	WP#	A15	I/O	MMC Write Protect(Active low)

	WP#		I/O	SD Write Protect(Active low)
	FGPIO[56]		I/O	General purpose I/O 56 shared with memory card
FMSINS	MS_INS	D14	I/O	MEMORY STICK PRO Insert
	FGPIO[57]		I/O	General purpose I/O 57 shared with memory card
FXDWPx	XWP#	A14	I/O	XD Write Protect (Active low)
	FGPIO[58]		I/O	General purpose I/O 58 shared with memory card
FXDCDx	XCD#	B14	I/O	XD Card Detect(Active low)
	FGPIO[59]		I/O	General purpose I/O 59 shared with memory card

3. UART Interface (2)

Pin Name	*Default*	*Alt. Fnuc. 1*	*Type*
URX	UGPIO[0]	URX	IO
UTX	UGPIO[1]	UTX	IO

Pin Name		Pin Number	I/O	Definition
URX	URX	B11	I/O	UART Receive signal
	UGPIO[0]		I/O	General purpose I/O 0 shared with UART
UTX	UTX	C11	I/O	UART Transmit signal
	UGPIO[1]		I/O	General purpose I/O 1 shared with UART

4. USB Interface (7)

Pin Name	*Default*	*Type*	
USBFDM	USBDM	Analog	FS
USBFDP	USBDP	Analog	FS
USBHDM	USBDM	Analog	HS
USBHDP	USBDP	Analog	HS
UREFEXT	UREFEXT	Analog	HS
URDATAP	URDATAP	Analog	HS
URDATAN	URDATAN	Analog	HS

Pin Name		Pin Number	I/O	Definition
USBFDM	USBDM	T13	ANALOG	USB Host full speed D- signal
USBFDP	USBDP	M13	ANALOG	USB Host full speed D+ signal
USBHDM	USBDM	R13	ANALOG	USB Device full speed D- signal
USBHDP	USBDP	N12	ANALOG	USB Device full speed D+ signal
UREFEXT	UREFEXT	N11	ANALOG	

URDATAP	URDATAP	N13	ANALOG	
URDATAN	URDATAN	P13	ANALOG	

5. Digital LCD Interface (20)

Pin Name	*Default*	*Alt. Func. 1*						*Alt. Func. 2*	*Alt. Func. 3*	*Type*
		UNIPAC	*EPSON*	*CCIR*	*CASIO*	*SONY/SANYO*	*Sharp Analog*			
VD0	VGPIO[0]	VD[0]	VD[0]	YC[0]	VD[0]	VD[0]	MOD[1]		VG[0]	IO
VD1	VGPIO[1]	VD[1]	VD[1]	YC[1]	VD[1]	VD[1]	MOD[2]		VG[1]	IO
VD2	VGPIO[2]	VD[2]	VD[2]	YC[2]	VD[2]	VD[2]	(SDAC -->)	VR[0]	VG[2]	IO
VD3	VGPIO[3]	VD[3]	VD[3]	YC[3]	VD[3]	VD[3]	(SDAC -->)	VR[1]	VG[3]	IO
VD4	VGPIO[4]	VD[4]	VD[4]	YC[4]	VD[4]	VD[4]	(SDAC -->)	VG[0]	VG[4]	IO
VD5	VGPIO[5]	VD[5]	VD[5]	YC[5]	VD[5]	VD[5]	(SDAC -->)	VG[1]	VG[5]	IO
VD6	VGPIO[6]	VD[6]	LP	YC[6]	GSRT	VD[6]	(SDAC -->)	VB[0]	VG[6]	IO
VD7	VGPIO[7]	VD[7]	RES	YC[7]	GRES	VD[7]	(SDAC -->)	VB[1]	VG[7]	IO
VDCK	VGPIO[8]	VDCK	(VDCK)	VDCK (27MHZ)	CLK (6.75MHZ)	MCLK (27MHZ)	(SDAC -->)	VPXCK	VPXCK	IO
VHSYNC	VGPIO[9]	VHSYNC	XSCL	VHSYNC	STBYB	XHD	CLD	DVHSYNC	DVHSYNC	IO
VDVALID	VGPIO[10]	VDVALID	FRYS	VDVALID			SPS	DVDVALID	DVDVALID	IO
VVSYNC	VGPIO[11]	VVSYNC	FRYP	VVSYNC	CP	XVD	SPOI	DVVSYNC	DVVSYNC	IO
VCCIR_C0	VGPIO[12]		GCP	VCCIR_C[0]	GPCK (14.5KHZ)		(SDAC -->)	VPX4CK	VB[0]	IO
VCCIR_C1	VGPIO[13]		FRX	VCCIR_C[1]			CLS	VY[0]	VB[1]	IO
VCCIR_C2	VGPIO[14]		YSCL	VCCIR_C[2]	STH		SPIO	VY[1]	VB[2]	IO
VCCIR_C3	VGPIO[15]		YSCLD	VCCIR_C[3]	STB		CTR	VC[0]	VB[3]	IO
VCCIR_C4	VGPIO[16]		DYIO2/DY	VCCIR_C[4]	POL		COM	VC[1]	VB[4]	IO
VCCIR_C5	VGPIO[17]			VCCIR_C[5]					VB[5]	IO
VCCIR_C6	VGPIO[18]			VCCIR_C[6]					VB[6]	IO
VCCIR_C7	VGPIO[19]			VCCIR_C[7]					VB[7]	IO
GP8	GP8								VR0	IO
GP9	GP9								VR1	IO
GP10	GP10								VR2	IO
GP11	GP11								VR3	IO
GP12	GP12								VR4	IO
GP13	GP13								VR5	IO
GP14	GP14								VR6	IO
GP15	GP15								VR7	IO

Notes: 1. For CASIO LCD COM16T1151 support, the following signals are controlled by GPIOs.
RIT Horizontally Flipping Control
2. Alt. Func. 2 is for Serial DAC output
3. Alt. Func. 3 is for 24-bit digital RGB, YCbCr or YPbPr output
==> Set VGPIO and GPIO[15:8] to Alt. Func. 3

Pin Name		Pin Number	I/O	Definition
VD[0]	VG[0]	N5	O	Digital Green / Y Output bit 0
	VGPIO[0]		I/O	General purpose I/O 0 Shared with Video Output pin

VD[1]	VG[1]	R3	O	Digital Green / Y Output bit 1
	VGPIO[1]		I/O	General purpose I/O 1 Shared with Video Output pin
VD[2]	VG[2]	T3	O	Digital Green / Y Output bit 2
	VGPIO[2]		I/O	General purpose I/O 2 Shared with Video Output pin
VD[3]	VG[3]	P6	O	Digital Green / Y Output bit 3
	VGPIO[3]		I/O	General purpose I/O 3 Shared with Video Output pin
VD[4]	VG[4]	T4	O	Digital Green / Y Output bit 4
	VGPIO[4]		I/O	General purpose I/O 4 Shared with Video Output pin
VD[5]	VG[5]	R4	O	Digital Green / Y Output bit 5
	VGPIO[5]		I/O	General purpose I/O 5 Shared with Video Output pin
VD[6]	VG[6]	N6	O	Digital Green / Y Output bit 6
	VGPIO[6]		I/O	General purpose I/O 6 Shared with Video Output pin
VD[7]	VG[7]	P7	O	Digital Green / Y Output bit 7
	VGPIO[7]		I/O	General purpose I/O 7 Shared with Video Output pin
VDCK	VPXCK	R5	O	Pixel Clock Output
	VGPIO[8]		I/O	General purpose I/O 7 Shared with Video Output pin
VHSYNC	DVHSYNC	N7	O	H-SYNC Output
	VGPIO[9]		I/O	General purpose I/O 7 Shared with Video Output pin
VDVALID	DVDVALID	T5	O	Data Output Valid (Enable)
	VGPIO[10]		I/O	General purpose I/O 7 Shared with Video Output pin
VVSYNC	DVVSYNC	P8	O	V-SYNC Output
	VGPIO[11]		I/O	General purpose I/O 7 Shared with Video Output pin
VCCIR_C0	VB[0]	R6	O	Digital Blue / CbCr/ Pb Output bit 0
	VGPIO[12]		I/O	General purpose I/O 12 Shared with Video Output pin
VCCIR_C1	VB[1]	T6	O	Digital Blue / CbCr/ Pb Output bit 1
	VGPIO[13]		I/O	General purpose I/O 13 Shared with Video Output pin
VCCIR_C2	VB[2]	R7	O	Digital Blue / CbCr/ Pb Output bit 2
	VGPIO[14]		I/O	General purpose I/O 14 Shared with Video Output pin
VCCIR_C3	VB[3]	N8	O	Digital Blue / CbCr/ Pb Output bit 3
	VGPIO[15]		I/O	General purpose I/O 15 Shared with Video Output pin
VCCIR_C4	VB[4]	T7	O	Digital Blue / CbCr/ Pb Output bit 4
	VGPIO[16]		I/O	General purpose I/O 16 Shared with Video Output pin
VCCIR_C5	VB[5]	R8	O	Digital Blue / CbCr/ Pb Output bit 5

	VGPIO[17]		I/O	General purpose I/O 17 Shared with Video Output pin
VCCIR_C6	VB[6]	T8	O	Digital Blue / CbCr/ Pb Output bit 6
	VGPIO[18]		I/O	General purpose I/O 18 Shared with Video Output pin
VCCIR_C7	VB[7]	P9	O	Digital Blue / CbCr/ Pb Output bit 7
	VGPIO[19]		I/O	General purpose I/O 19 Shared with Video Output pin
GP8	VR[0]	T9	I/O	Digital Red / Pr Output bit 0 shared with GPIO[8]
GP9	VR[1]	N9	I/O	Digital Red / Pr Output bit 0 shared with GPIO[9]
GP10	VR[2]	R9	I/O	Digital Red / Pr Output bit 0 shared with GPIO[10]
GP11	VR[3]	R10	I/O	Digital Red / Pr Output bit 0 shared with GPIO[11]
GP12	VR[4]	T10	I/O	Digital Red / Pr Output bit 0 shared with GPIO[12]
GP13	VR[5]	R11	I/O	Digital Red / Pr Output bit 0 shared with GPIO[13]
GP14	VR[6]	P11	I/O	Digital Red / Pr Output bit 0 shared with GPIO[14]
GP15	VR[7]	T11	I/O	Digital Red / Pr Output bit 0 shared with GPIO[15]

6. VDAC Interface (5)

Pin Name	*Default*	*Type*
VIROUT	VIROUT	Analog
VIGOUT	VIGOUT	Analog
VIBOUT	VIBOUT	Analog
VIREF	VIREF	Analog
VCOMPA	VCOMPA	Analog

Pin Name		Pin Number	I/O	Definition
AVIROUT	VIROUT	T2	ANALOG	Analog Red/Cr/Pr channel output
AVIGOUT	VIGOUT	P5	ANALOG	Analog Green/Y channel output
AVIBOUT	VIBOUT	T1	ANALOG	Analog Blue channel output
VIREF	VIREF	N4	ANALOG	Reference Voltage
VCOMPA	VCOMPA	P4	ANALOG	Voltage Compensted

7. GPIO Interface (32)

Pin Name	*Default*	*Alt. Func. 1*	*Alt. Func. 2*	*Alt. Func. 3*	*Type*
GP0	GP0 PLL2XIN/ USBCKIN/ AUXCKI	PWMQ0	UDATA0		IO
GP1	GP1 MEMCKIN	PWMQ1	UDATA1		IO

GP2	GP2 CPUCKIN	PWMQ2	UDATA2		IO	
GP3	GP3	PWMQ3	UDATA3		IO	
GP4	GP4 /SPIMI0 (I)	SPISO0 (O)	UDATA4		IO	
GP5	GP5 /SPISI0 (I)	SPIMO0 (O)	UDATA5		IO	
GP6	GP6/ SPICS0# (I)	SPICS0# (O)	UDATA6		IO	
GP7	GP7/ SPISCK0 (I)	SPISCK0 (O)	UDATA7		IO	
GP8	GP8	MDMAD[0]	UDATA8	VR0	IO	
GP9	GP9	MDMAD[1]	UDATA9	VR1	IO	
GP10	GP10	MDMAD[2]	UDATA10	VR2	IO	
GP11	GP11	MDMAD[3]	UDATA11	VR3	IO	
GP12	GP12	MDMAD[4]	UDATA12	VR4	IO	
GP13	GP13	MDMAD[5]	UDATA13	VR5	IO	
GP14	GP14	MDMAD[6]	UDATA14	VR6	IO	
GP15	GP15	MDMAD[7]	UDATA15	VR7	IO	
GP16	GP16/ SPISI1 (I)/ IRDIN	SPIMO1 (O)	UVALIDH		IO	IRDIN is the IR data input
GP17	GP17/ SPICS1# (I)	SPICS1# (O)	UTXVALID		IO	
GP18	GP18/ SPISCK1 (I)	SPISCK1 (O)	UTXRDY		IO	
GP19	GP19 /SPIMI1 (I)	SPISO1 (O)	URXVALID		IO	
GP20	GP20/ TM0TRG	SMCKO	URXACTV		IO	SMCKO and SMDQ are for I2C master
GP21	GP21/ TM1TRG	SMDQ	URXERROR		IO	
GP22	GP22/ TM2TRG/ SSCKI		USIECLK		IO	SSCKI and SSDQ are for I2C slave
GP23	GP23/ TM3TRG	SSDQ	USUSPNDx		IO	
GP24	GP24		URSTx	SDIO[0]	IO	
GP25	GP25	MXWAIT#	UOPMODE0	SDIO[1]	IO	
GP26	GP26	MXCS#	UOPMODE1	SDIO[2]	IO	
GP27	GP27	MDMAWR	ULNSTAT0	SDIO[3]	IO	
GP28	GP28	MDMARD	ULNSTAT1		IO	
GP29	GP29	MDMREQ	UXVRSEL		IO	
GP30	GP30	MDMACK	UTERMSEL			
GP31	GP31	ALE				

Note: Among all the GPIOs, only GP[31:0] .

Pin Name	Default	Pin Number	I/O	Definition
GP0	GP0	B13	I/O	Dedicated General purpose I/O 0
	PWMQ[0]		O	Pulse width Modulated Output 0

	UDATA[0]		I/O	Data bus 0
GP1	GP1	E11	I/O	Dedicated General purpose I/O 1
	PWMQ[1]		O	Pulse width Modulated Output 1
	UDATA[1]		I/O	Data bus 1
GP2	GP2	A13	I/O	Dedicated General purpose I/O 2
	PWMQ[2]		O	Pulse width Modulated Output 2
	UDATA[2]		I/O	Data bus 2
GP3	GP3	A12	I/O	Dedicated General purpose I/O 3
	PWMQ[3]		O	Pulse width Modulated Output 3
	UDATA[3]		I/O	Data bus 3
GP4	GP4	C12	I/O	Dedicated General purpose I/O 4
	UDATA[4]		I/O	Data bus 4
GP5	GP5	D11	I/O	Dedicated General purpose I/O 5
	UDATA[5]		I/O	Data bus 5
GP6	GP6	B12	I/O	Dedicated General purpose I/O 6
	UDATA[6]		I/O	Data bus 6
GP7	GP7	F11	I/O	Dedicated General purpose I/O 7
	UDATA[7]		I/O	Data bus 7
GP8	GP8	T9	I/O	Dedicated General purpose I/O 8
	MDMAD[0]		I/O	DMA Data bus 0
	UDATA[8]		I/O	Data bus 8
	VR[0]		O	Digital Red / Pr Output bit 0 shared with GPIO[8]
GP9	GP9	N9	I/O	Dedicated General purpose I/O 9
	MDMAD[1]		I/O	DMA Data bus 1
	UDATA[9]		I/O	Data bus 9
	VR[1]		O	Digital Red / Pr Output bit1 shared with GPIO[9]
GP10	GP10	R9	I/O	Dedicated General purpose I/O 10
	MDMAD[2]		I/O	DMA Data bus 2
	UDATA[10]		I/O	Data bus 10
	VR[2]		O	Digital Red / Pr Output bit 2 shared with GPIO[10]
GP11	GP11	R10	I/O	Dedicated General purpose I/O 11
	MDMAD[3]		I/O	DMA Data bus 3
	UDATA[11]		I/O	Data bus 11

	VR[3]		O	Digital Red / Pr Output bit 3 shared with GPIO[11]
GP12	GP12	T10	I/O	Dedicated General purpose I/O 12
	MDMAD[4]		I/O	DMA Data bus 4
	UDATA[12]		I/O	Data bus 12
	VR[4]		O	Digital Red / Pr Output bit 4 shared with GPIO[12]
GP13	GP13	R11	I/O	Dedicated General purpose I/O 13
	MDMAD[5]		I/O	DMA Data bus 5
	UDATA[13]		I/O	Data bus 13
	VR[5]		O	Digital Red / Pr Output bit 5 shared with GPIO[13]
GP14	GP14	P11	I/O	Dedicated General purpose I/O 14
	MDMAD[6]		I/O	DMA Data bus 6
	UDATA[14]		I/O	Data bus 14
	VR[6]		O	Digital Red / Pr Output bit 6 shared with GPIO[14]
GP15	GP15	T11	I/O	Dedicated General purpose I/O 15
	MDMAD[7]		I/O	DMA Data bus 7
	UDATA[15]		I/O	Data bus 15
	VR[7]		O	Digital Red / Pr Output bit 7 shared with GPIO[15]
GP16	GP16	A11	I/O	Dedicated General purpose I/O 16
	IRDIN		I	Infra Red signal data input
	UVALIDH		I/O	
GP17	GP17	D10	I/O	Dedicated General purpose I/O 17
	UTXVALID		I/O	
GP18	GP18	A10	I/O	Dedicated General purpose I/O 18
	UTXRDY		I/O	
GP19	GP19	E10	I/O	Dedicated General purpose I/O 19
	URXVALID		I/O	
GP20	GP20	B10	I/O	Dedicated General purpose I/O 20
	SMCKO		I/O	I2C interface clock pin (Master mode)
	URXACTV		I/O	
GP21	GP21	D7	I/O	Dedicated General purpose I/O 21
	SMDQ		I/O	I2C interface data pin (Master mode)
	URXERROR		I/O	
GP22	GP22	A9	I/O	Dedicated General purpose I/O 22

	USIECLK		I/O	I2C interface clock pin (Slave mode)
GP23	GP23	C8	I/O	Dedicated General purpose I/O 23
	SSDQ		I/O	I2C interface data pin (Slave mode)
	USUSPNDx		I/O	
GP24	GP24	A8	I/O	Dedicated General purpose I/O 24
	URSTx		I/O	
	SDIO[0]		I/O	SDIO Date bus 0
GP25	GP25	D9	I/O	Dedicated General purpose I/O 25
	MXWAIT#		I/O	
	UOPMODE[0]		I/O	
	SDIO[1]		I/O	SDIO Date bus 1
GP26	GP26	C10	I/O	Dedicated General purpose I/O 26
	MXCS#		I/O	
	UOPMODE[1]		I/O	
	SDIO[2]		I/O	SDIO Date bus 2
GP27	GP27	E9	I/O	Dedicated General purpose I/O 27
	MDMAWR		I/O	DMA Write
	ULNSTAT[0]		I/O	
	SDIO[3]		I/O	SDIO Date bus 3
GP28	GP28	B9	I/O	Dedicated General purpose I/O 28
	MDMARD		I/O	DMA Read
	ULNSTAT[1]		I/O	
GP29	GP29	C6	I/O	Dedicated General purpose I/O 29
	MDMREQ		I/O	DMA Request
	MXVRSEL		I/O	
GP30	GP30	A7	I/O	Dedicated General purpose I/O 30
	MDMACK		I/O	DMA Clock
	MTERMSEL		I/O	
GP31	GP31	B8	I/O	Dedicated General purpose I/O 31
	ALE		I/O	DMA Address Latch Enable

8. Audio Interface (5)

Pin Name	Default	Alt. Func. 1	Type
ACLK	AGPIO[0]	ACLK	IO
AMCLK	AGPIO[1]	AMCLK/ ARSYNC	IO
AFSYNC	AGPIO[2]	AFSYNC/ ATSYNC	IO
ADIN	AGPIO[3]	ADIN	IO
ADOUT	AGPIO[4]	ADOUT	IO

Pin Name		Pin Number	I/O	Definition
ACLK	ACLK	D13	I/O	Acknowledge signal
AMCLK	AMCLK/ ARSYNC	C14	I/O	
AFSYNC	AFSYNC/ ATSYNC	D12	I/O	
ADIN	ADIN	C13	I/O	
ADOUT	ADOUT	E12	I/O	

9. RTC Interface (3)

Pin Name	Default	Type
RXIN	RXIN	I
RXOUT	RXOUT	O
RALARM	RALARM	O

Pin Name		Pin Number	I/O	Definition
RXIN	RXIN	P12	I	
RXOUT	RXOUT	N10	O	
RALARM	RALARM	T12	O	

10. Miscellaneous (9)

Pin Name	Default	Type	
RESETx	RESETx	I	
RSTOUTx	RSTOUTx	O	
CLKIN	CLKIN	I	
CLKOUT	CLKOUT	O	
NAFBOOT	NAFBOOT	I	Boot up from NAND Flash
TESTMD[0]	TESTMD[0]	I	
TESTMD[1]	TESTMD[1]	I	
TESTMD[2]	TESTMD[2]	I	
SCANEN	SCANEN	I	

Pin Name		Pin Number	I/O	Definition
RESETx	RESETx	A6	I	
RSTOUTx	RSTOUTx	C7	O	
CLKIN	CLKIN	C1	I	
CLKOUT	CLKOUT	C2	O	
NAFBOOT	NAFBOOT	C9	I	
TESTMD[0]	TESTMD[0]	B7	I	
TESTMD[1]	TESTMD[1]	A5	I	
TESTMD[2]	TESTMD[2]	F4	I	
SCANEN	SCANEN	B6	I	

8 Electrical Characteristics

8.1 Absolute Maximum Ratings

Symbol	Parameter	Value	Units
VPP	I/O supply Voltage (DVPP, AVPP1, AVPP2, UVPP)	-1.0 to 4.6	V
VDD	Core supply Voltage (DVDD, AVDD1, AVDD2)	-1.0 to 4.6	V
Vin5	DC Input Voltage for 5V-tolerant I/O [2]	-1.0 to 5.5	V
Vin	DC Input Voltage for non-5V-tolerant I/O	-1.0 to 4.6	V
TSTG	Storage Temperature	-40 to 125	°C
ESD	ESD Rating (Rzap = 1.5K Ω, Czap = 100pf)	TBD	V

Note: 1. Permanent device damage may occurs if the specification for the Absolute Maximum Ratings are exceeded.

2. 5V tolerant I/O including URX, UTX, GP0, GP1 and GP[15:8].

3. All voltages are defined with respect to ground.

8.2 Recommended Operating Conditions

Symbol	Parameter	Min.	Typ.	Max.	Units
VPP	I/O supply Voltage (DVPP, AVPP1, AVPP2, UVPP)	3.1	3.3	3.6	V
VDD	Core supply Voltage (DVDD, AVDD1, AVDD2)	1.68	1.8	1.98	V
TA	Ambient Operating Temperature	0	-	70	°C

8.3 DC Characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Units
V_{IL}	Input Low Voltage	TTL			0.8	V
V_{IH}	Input High Voltage	TTL	2.2			V
V_{IL} –SCH [1]	Schmitt-triggered Input Low Voltage	Schmitt Trig. TTL			0.9	V
V_{IH} –SCH [1]	Schmitt trig. Input High Voltage	Schmitt Trig. TTL	1.9			V
V_{OL}	Output Low Voltage	4mA load			0.4	V
V_{OH}	Output High Voltage	4mA load	0.8*VPP			V

R_d [2]	Pull Down Resistance	-	91	120	215	KΩ
I_i	Input Leakage Current	Vo = 3.3V or 0V	-	10nA	1μA	
Ioz	Tri-state Output Leakage Current	Vo = 3.3V or 0V		10nA	1μA	

Note: 1. The I/O with Schmitt-trigger: TESTMD, RESETx and GP[1:0].

2. The I/O with pull-down: TESTMD.

8.4 Capacitance

Symbol	Parameter	Min.	Typ.	Max.	Units
C_{XIN}	Clock Input, CLKIN, capacitance		7		pF
C_{IN}	Input pin capacitance		5		pF
C_{BI}	Bidirectional pin capacitance		5		pF
C_{BI5}	Bidirectional pin capacitance for 5V-tolerant I/O		7		pF
C_{OUT}	Output pin capacitance		5		pF

8.5 AC Characteristics

8.5.1 *Reset Timing*

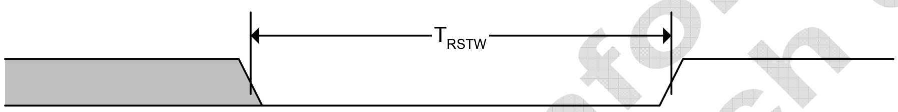
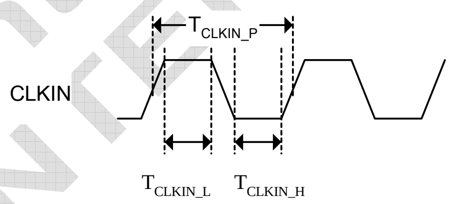


Fig. 8-1 Reset Timing

Symbol	Parameter	Min.	Typ.	Max.	Unit
T_{RSTW}	Reset Pulse Width	1			ms

8.5.2 *Input Clock*

Fig. 8-2 CLKIN AC Characteristic



Symbol	Parameter	Min.	Typ.	Max.	Unit
F_{CLKIN}	CLKIN Master Clock Input Frequency		13.5		MHZ
T_{CLKIN_P}	CLKIN Master Clock Input Period		74		ns
T_{CLKIN_L}	CLKIN Master Clock Input Low Width	33.3		40.7	ns
T_{CLKIN_H}	CLKIN Master Clock Input High Width	33.3		40.7	ns
DT_{CLKIN}	CLKIN Duty Cycle	45		55	%

8.5.3 *SDRAM Interface Timing*

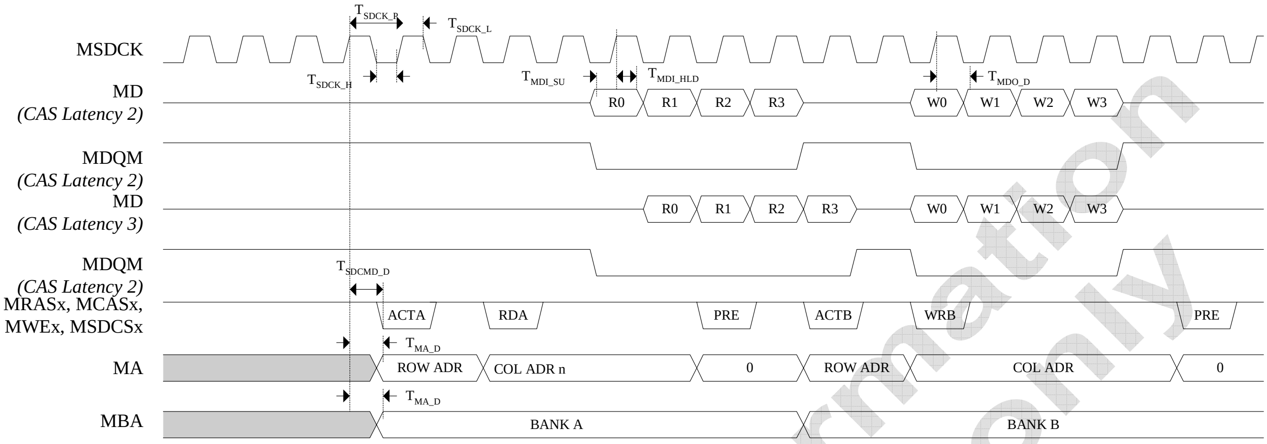


Fig. 8-3 Basic SDRAM Interface Timing

Symbol	Parameter	Min.	Typ.	Max.	Unit
T_{SDCK_P}	SDRAM Clock Output Period	13.9			ns
T_{SDCK_H}	SDRAM Clock Output High Pulse Width	4.6			ns
T_{SDCK_L}	SDRAM Clock Output Low Pulse Width	4.6			ns
T_{MDI_SU}	SDRAM Data Input Setup Time	TBD			ns
T_{MDI_HLD}	SDRAM Data Input Hold Time	TBD			ns
T_{MA_D}	SDRAM Address Valid Delay			TBD	ns
T_{MDO_D}	SDRAM Data Output Valid Delay			TBD	ns
T_{SDCMD_D}	SDRAM Command Valid Delay			TBD	ns

8.5.4 *Display Output*

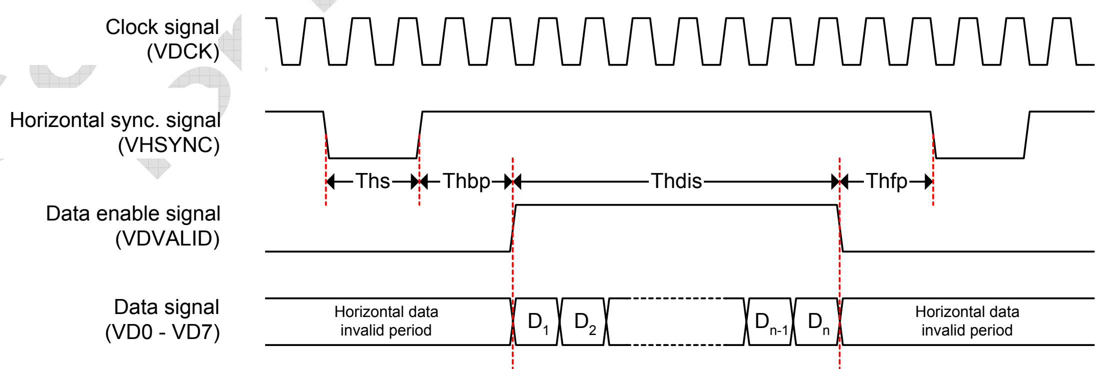


Fig. 8-4 Display/CCIR-601 Output Horizontal Timing

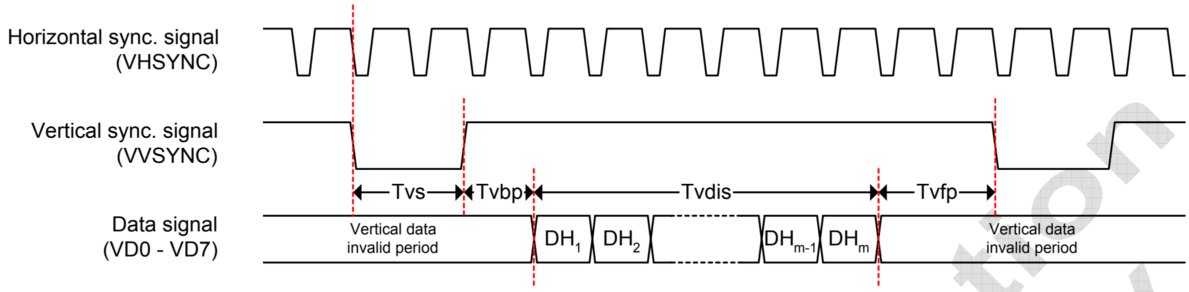


Fig. 8-5 Progressive Display Output Vertical Timing

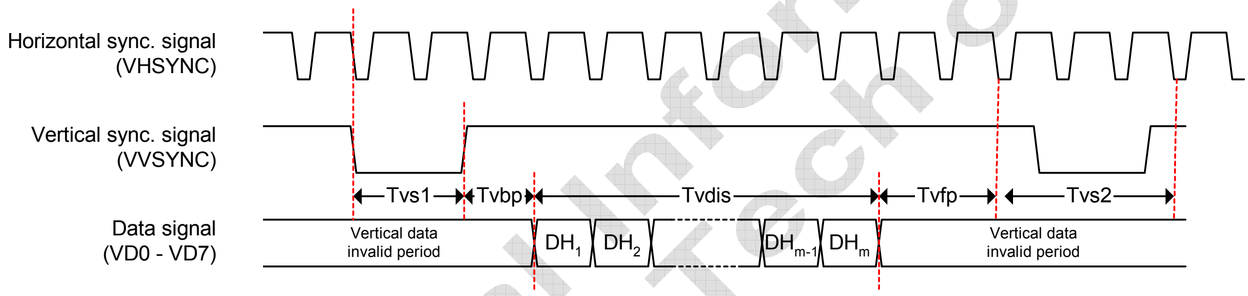


Fig. 8-6 Progressive Display Output Vertical Timing

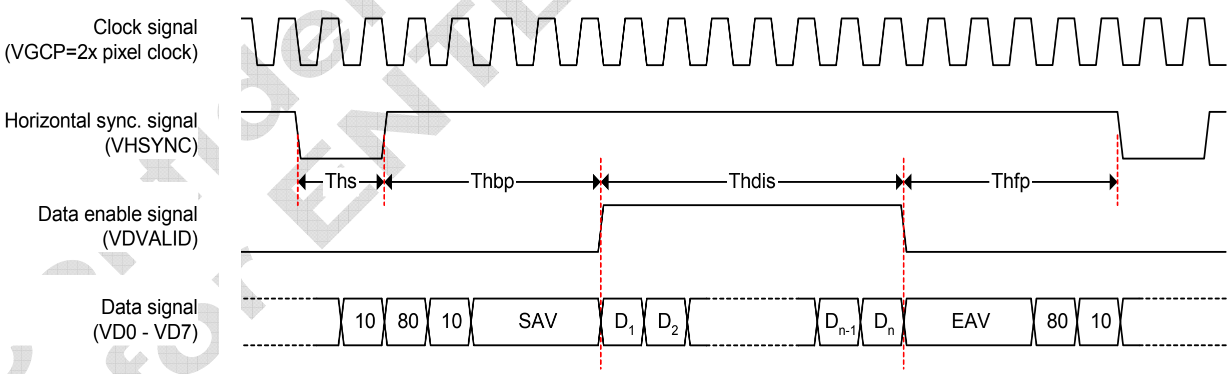


Fig. 8-7 CCIR-656 Output Horizontal Timing

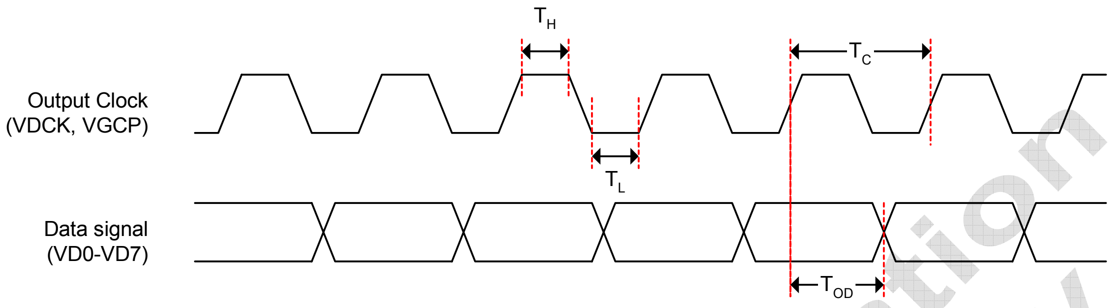
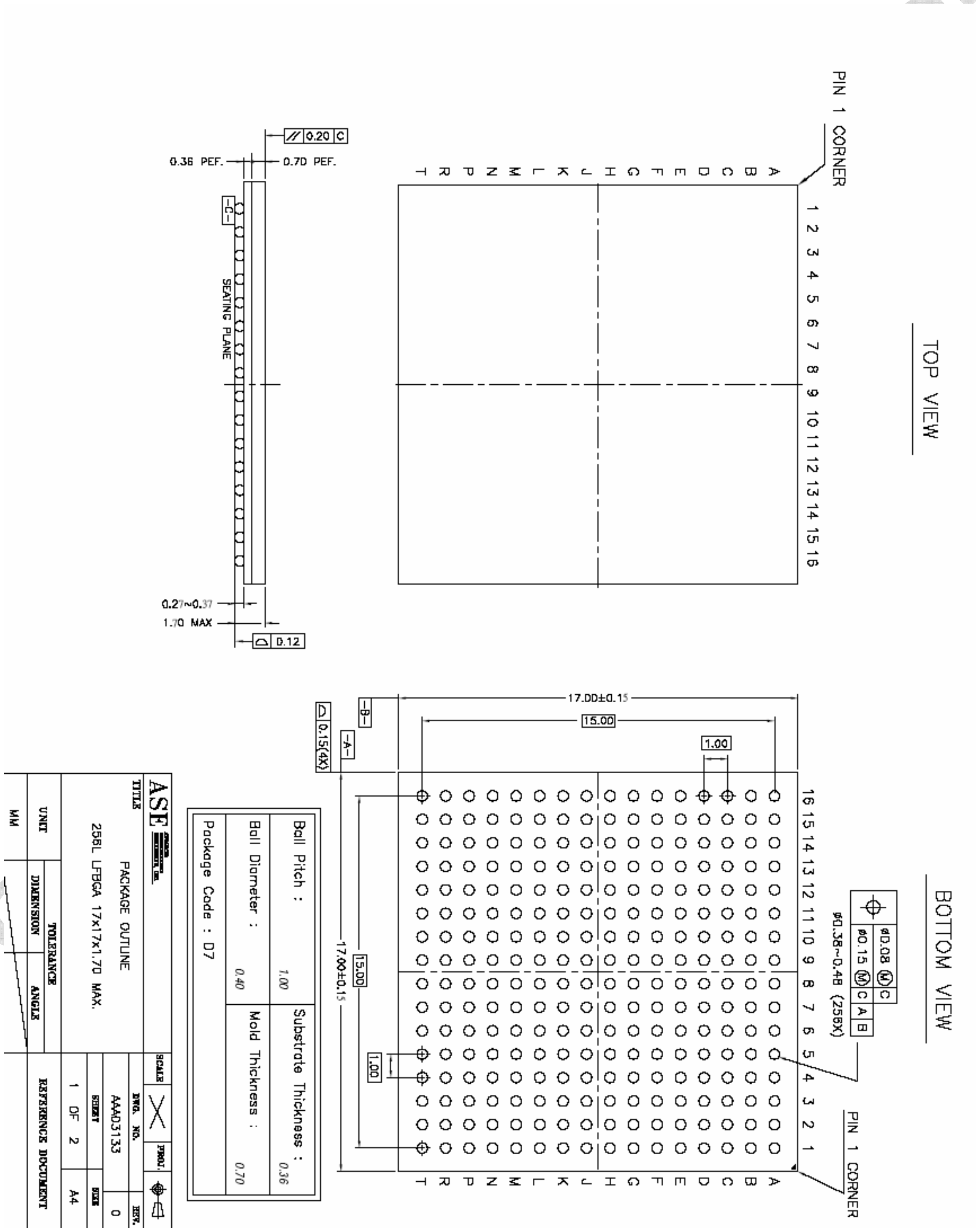


Fig. 8-8 Display Output Clock, and Data Sample Timing

Symbol		Parameter	Min.	Typ.	Max.	Unit
T_{hs}		Horizontal Sync. Pulse Width	1		256	VDCK
T_{hbp}		Display Horizontal Back Porch Width	1		256	VDCK
T_{hdis}		Horizontal Display Enable Width	1		2048	VDCK
T_{hfp}		Display Horizontal Front Porch Width	1		256	VDCK
T_{vs}		Vertical Sync. Pulse Width	1		256	line
T_{vbp}		Display Vertical Back Porch Width	1		256	line
T_{vdis}		Vertical Display Enable Width	1		2048	line
T_{vfp}		Display Vertical Back Porch Width	1		256	line
T_C	VDCK	VDCK Clock Cycle Time	65			ns
	VGCP	VGCP (2x of VDCK) Clock Cycle Time	33			ns
T_H	VDCK	VDCK Clock High Pulse Width	20			ns
	VGCP	VGCP (2x of VDCK) Clock High Pulse Width	11			ns
T_L	VDCK	VDCK Clock Low Pulse Width	20			ns
	VGCP	VGCP (2x of VDCK) Clock Low Pulse Width	11			ns
T_{OD}	VDCK	Display Data Output delay time relative to the rising edge of VDCK			TBD	ns
	VGCP	Display Data Output delay time relative to the rising edge of VGCP			TBD	ns

8.6 Package Dimensions

Package Outline for LFBGA 256 Ball (17mm x 17mm x 1.6mm, Ball Pitch = 1.0mm, Ball Diameter = 0.5mm)



9 Revision History

Date	Revision	Description
Nov. 09. 2004	0.7	Initial Version
Mar. 24. 2005	0.8	Added Ball Assignment
Mar. 30. 2005	0.81	Change Ball Assignment (Top View)